# Adjective Ordering in Referential Communication: Subjectivity, Discriminatory Strength, and Incremental Pragmatics

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**Abstract**

Adjective ordering preferences are robust across languages, yet speakers also adapt adjective order to the referential context. This paper asks how stable ordering expectations, context-dependent informativeness, and incremental choice jointly shape adjective order in referential communication. We develop a Rational Speech Act model that crosses speaker architecture (global vs. incremental) with semantic regime (context-fixed vs. context-updating), and evaluate the same model space against two German referential experiments: a slider-rating study and a constrained-production study. Across both tasks, order preferences vary systematically with the discriminatory value of the adjectives in the current scene while showing a baseline size-first tendency. Bayesian model comparison shows a clear architecture effect: incremental speakers outperform global speakers in both tasks. By contrast, semantic regime plays a limited role in the fitted models: context-updating and context-fixed semantics are indistinguishable for slider judgments, and context-fixed semantics is slightly favored in production. The production experiment further reveals systematic overspecification, with redundant use of modifiers increasing when size is harder to discriminate; the production model captures this pattern by adding a graded visual-salience layer on top of pragmatic informativeness, allowing perceptual prominence to shape both adjective ordering and selection. Simulation shows that context-updating semantics can nevertheless generate a baseline size-first asymmetry on its own. The results support a multi-level account in which adjective order reflects the interaction of stable ordering expectations, context-sensitive informativeness, perceptual salience, and incremental decision-making. This account connects subjectivity-based explanations of canonical order with efficiency-based accounts of reference while clarifying which mechanisms are needed to predict behavior in judgment and production tasks.

*Keywords:* adjective ordering, Rational Speech Act, incremental pragmatics, communicative efficiency, subjectivity

**Adjective Ordering in Referential Communication:**
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## 1 Introduction

Adjective ordering provides a window onto how speakers coordinate stable linguistic preferences with the demands of situated reference. In a phrase such as *the big blue sticker*, the speaker chooses which properties to mention and the sequence in which those properties will guide the listener's search. This sequence is constrained by robust ordering preferences: across languages, speakers tend to prefer orders such as "big brown bag" over "brown big bag" (Scontras, 2023; Sproat & Shih, 1991). At the same time, the referential scene creates local pressures on ordering. If colour identifies the target more efficiently than size, the communicatively useful first modifier may conflict with the canonical size-first preference (Fukumura, 2018; Rubio-Fernández, Mollica, & Jara-Ettinger, 2021).

These observations raise a mechanistic question for theories of adjective ordering. Stable semantic or lexical factors account for why some adjective classes tend to occur closer to the noun, while context-sensitive efficiency accounts explain why locally useful properties may be mentioned early. What remains underspecified is how these sources of order preference are combined, and which component of the speaker-listener system carries the context sensitivity. In particular, context sensitivity can enter through semantic composition, when the interpretation of a gradable adjective changes as the comparison class is narrowed, or through speaker choice, when the speaker selects modifiers one at a time in response to the current discriminatory value of each property.

We ask whether context-sensitive adjective ordering is better explained by context-updating semantic composition, by incremental speaker choice, or by both mechanisms together, once stable order expectations and perceptual salience are represented explicitly. If context sensitivity comes from semantic composition, adjective meanings become order-dependent because each modifier is interpreted relative to the comparison class established during composition. If it comes from production, meanings can remain fixed relative to the scene while the speaker's word-by-word choices adapt to the listener's evolving information state.

We address this question in three linked steps. First, the experiments establish which qualitative ordering patterns the model aims to explain: a stable size-first pressure, modulation by referential context, and the richer length and redundancy patterns revealed in production. Second, the model comparison asks which mechanisms improve out-of-sample prediction of these patterns once they are placed in the same RSA architecture. Third, the simulation asks what context-updating composition contributes as a generative mechanism, independently of fitted production parameters.

To implement this comparison, we develop a Rational Speech Act (RSA) model (Degen, 2023; Goodman & Frank, 2016) that crosses two dimensions in a $2 \times 2$ design: *speaker architecture* (global vs. incremental) and *semantic regime* (context-fixed vs. context-updating). The global speaker evaluates complete utterances; the incremental speaker chooses one modifier at a time. Under context-updating semantics, the interpretation of an outer adjective such as "big" depends on the comparison class established by the material it composes with, so that "big blue" and "blue big" need not be equivalent; under context-fixed semantics, the size threshold is determined once from the scene. All four model variants are fitted to data from two complementary referential experiments in German — a slider-rating study measuring graded ordering preferences and a constrained production study measuring actual utterance choice — in a shared paradigm that varies discriminatory strength across scenes. Using the same formal model space across the two datasets allows the comparison to function as a cross-task test of the proposed mechanisms: the same model variants are evaluated against both explicit order ratings and categorical production choices. Model comparison uses Bayesian leave-one-out cross-validation (PSIS-LOO; Vehtari, Gelman, & Gabry, 2017), which quantifies how much predictive accuracy each mechanism adds. A simulation study applies the same $2 \times 2$ crossing outside the fitted empirical models, asking which ordering effects each mechanism can generate on its own and in combination.

Taken together, the experiments, model comparison, and simulation support three conclusions. Across the two experiments, ordering preferences vary systematically with discriminatory strength: speakers tend to place the more informative adjective earlier, while the canonical size-first preference remains visible as a defeasible baseline. The model comparison shows that predictive fit is driven mainly by speaker architecture: incremental speakers outperform global speakers in both tasks, whereas the semantic-regime contrast is small or absent within the best-fitting incremental models. However, the simulation shows that context-updating composition can still generate a size-first advantage without incremental speaker choice; under context-fixed semantics, a global speaker has no corresponding ordering advantage. Figure 1 provides a schematic overview of the modelling framework, experiments, and key findings.

The remainder of the paper is structured as follows. Section 2 reviews prior work on adjective ordering, referential efficiency, and incremental pragmatic modelling. Section 3 introduces the modelling framework, and Section 5 gives the formal specification. Section 4 reports the two referential studies. Sections 6 and 7 present the fitted model comparison and simulation study. Section 8 summarizes the results and discusses their implications.

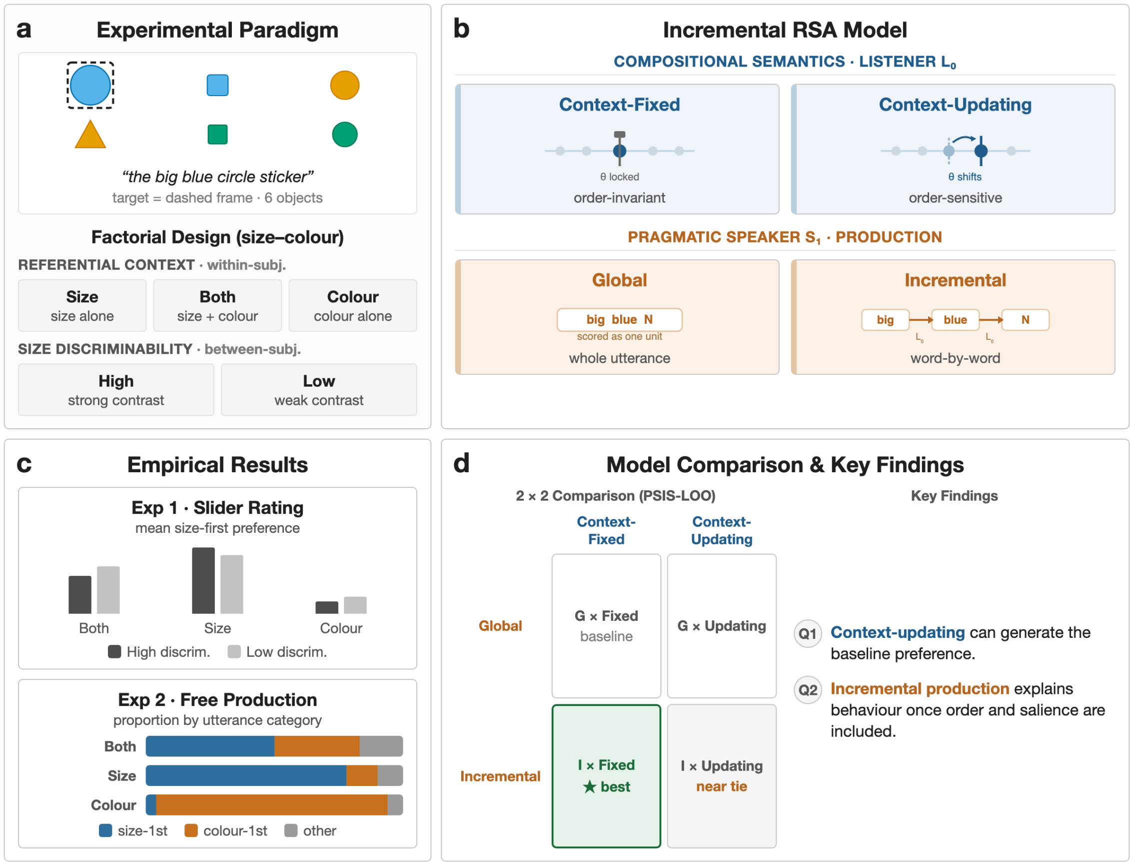



**Figure 1**

*Overview of the study: experimental paradigm, crossed RSA model space, empirical patterns, and* $2 \times 2$ *model comparison.*

## 2 Background

Research on adjective ordering has identified several stable predictors of preferred order, including meaning-based factors such as closeness to the noun and inherentness (Martin, 1969; Sweet, 1898; Whorf, 1945), class-based hierarchies (Cinque, 1994; Dixon, 1982; Quirk, Greenbaum, Leech, & Svartvik, 1972; Scott, 2002), and information-theoretic measures (Dyer, Torres, Scontras, & Futrell, 2023; Futrell, Dyer, & Scontras, 2020; Hahn, Degen, Goodman, Jurafsky, & Futrell, 2018). These predictors explain substantial structure, but residual variation remains even for strong individual predictors (for an overview, see Dyer et al., 2023; Scontras, 2023). Ordering preferences are also context-sensitive in ways that fixed hierarchies do not fully capture (Fukumura, 2018; Trotzke & Wittenberg, 2019). The present account therefore draws on three lines of work that speak directly to the source of this context sensitivity: subjectivity-based composition, referential efficiency, and incremental pragmatic choice.

### 2.1 Subjectivity and Adjective Ordering

A central empirical generalization in the adjective-ordering literature is the relation between an adjective's *subjectivity* and its preferred distance from the noun. Scontras, Degen, and Goodman (2017) operationalized subjectivity as susceptibility to faultless disagreement and showed that it explains a large share of variance in English ordering preferences, outperforming earlier predictors such as inherentness and intersectivity. Subsequent cross-linguistic work suggests that this relationship is not peculiar to English and may reflect a recurring regularity in adjective ordering (Scontras, 2023; Trainin & Shetreet, 2021). This relation is directly relevant to size–colour sequences because size adjectives are typically more subjective and gradable than colour adjectives, yielding an expected size-first baseline in prenominal modification.

Scontras, Degen, and Goodman (2019) argue that the explanatory core of this effect lies in restrictive composition. Assuming a hierarchical prenominal structure, the adjective closest to the noun composes first with the nominal denotation, and the outer adjective applies to the result of that restriction. The outer, more subjective adjective is therefore evaluated against a comparison class that has already been narrowed by the inner, less subjective adjective. In such a restricted comparison class, placing the less subjective adjective closer to the noun is advantageous because its interpretation provides a more stable basis for subsequent gradable modification. Franke, Scontras, and Simonic (2019) make the same intuition explicit in simulation, showing that subjectivity-aligned orders increase average communicative success under sequential composition with noisy meanings. For the present question, this explanation locates the ordering pressure in the compositional system rather than in a word-by-word production policy.

Scontras, Bar-Sever, Kachakeche, Rosales Jr, and Samonte (2020) develop this idea as *incremental semantic restriction*. If nominal structure supports sequential narrowing of the comparison class, then subjectivity-based ordering pressures should arise even without an incremental speaker in the usual process-model sense. This identifies a plausible compositional source for canonical ordering in the present paradigm: context-sensitive restrictive composition can favor placing the more subjective, gradable size adjective outside the less subjective colour adjective.

## 2.2 Communicative Efficiency and Discriminatory Strength

A second source of ordering pressure comes from referential efficiency. A separate line of work has shown that adjective use is sensitive to the demands of the immediate communicative context, including whether a modifier helps the listener identify the referent efficiently (Degen, Hawkins, Graf, Kreiss, & Goodman, 2020; Rubio-Fernández, 2016; Rubio-Fernandez, 2019). This literature treats adjective choice and adjective order as part of a broader problem of reference generation rather than as the expression of a static lexical hierarchy; a related commitment appears in Van Gompel and colleagues' conceptualization model, which derives property choice from the contrast set available to the speaker (Fukumura, Van Gompel, & Pickering, 2010; Van Gompel, Van Deemter, Gatt, Snoeren, & Krahmer, 2019).

Fukumura (2018) provide a direct empirical precedent for the present experiments. In their production studies, speakers were more likely to place the more discriminating adjective first, consistent with a *discriminatory efficiency hypothesis*: order should reflect which property best distinguishes the target in the current scene. The relevant pragmatic weight is therefore computed from the contrast set on the current trial. The same adjective may be highly useful in one array and much less useful in another.

Rubio-Fernández and colleagues develop this idea within a broader account of *incremental efficiency* (Rubio-Fernandez & Jara-Ettinger, 2020; Rubio-Fernández et al., 2021). They argue that the efficiency of a referential expression should be evaluated as the utterance unfolds, before the full description has been heard. This production-side view is closely aligned with incremental interpretation work showing that listeners use visual context and prenominal modifiers to restrict the candidate set before the noun is complete (Sedivy, Tanenhaus, Chambers, & Carlson, 1999; Tanenhaus, Spivey-Knowlton, Eberhard, & Sedivy, 1995). From this perspective, word order matters because it determines how the listener's search space is narrowed over time. Prenominal and postnominal systems therefore create different pragmatic affordances, and even redundant modifiers can be useful if they facilitate early search. The relevant quantity is the discriminatory value a property has at the stage of processing where it is encountered.

This work motivates treating discriminatory strength as a source of graded context sensitivity. The relevant test is whether size-first preferences persist, weaken, or reverse as size and colour differ in their usefulness for identifying the target, including when the size contrast itself is easier or harder to discriminate.

### 2.3 Incremental Pragmatic Models

The Rational Speech Act (RSA) framework provides the formal basis for comparing these pressures. In RSA, a pragmatic speaker selects utterances by reasoning about a listener's posterior, and the listener in turn reasons about the speaker's choice (Degen, 2023; Goodman & Frank, 2016). Although standard RSA treats complete utterances as the units of choice, the same logic can be defined incrementally, with the speaker choosing one word at a time on the basis of how each token updates the listener's beliefs (CohnGordon, Goodman, & Potts, 2019). Existing incremental pragmatic models and related work on redundancy show that word-by-word reasoning can capture asymmetries between adjective positions and between prenominal and postnominal systems (Degen et al., 2020; Rubio-Fernández et al., 2021). These accounts typically update the listener's posterior online while keeping adjective meanings fixed relative to the scene. Schlotterbeck and Wang (2023) introduced a closely related alternative, together with the slider-rating paradigm that forms the basis for one of the experiments reported here. In that model, the threshold for a gradable adjective is recomputed from the listener's current posterior at each stage of composition. The study provided a qualitative fit in which the model captured both a subjectivity-driven baseline ordering and a discriminatory-strength modulation. It did not separate context-updating semantic composition from incremental speaker choice in a joint quantitative comparison, so the same qualitative pattern remained compatible with multiple mechanisms.

# 3 The Modelling Framework

We extend Schlotterbeck and Wang (2023) by separating the mechanisms left open by the preceding literature. The framework embeds their proposal in a systematic $2 \times 2$ comparison of speaker architecture and semantic regime, including a context-fixed ablation. It adds a constrained production experiment with a richer response space and a simulation study that asks which ordering effects follow from context-updating composition outside the fitted empirical models. Bayesian model comparison then quantifies which mechanisms add predictive accuracy once production-specific order and salience pressures are represented explicitly. Here we describe the framework at a conceptual level; the meaning functions, speaker definitions, and inference setup are given formally in Section 5. The remaining subsections develop these components in turn.

## 3.1 Reference, Scenes, and the Speaker's Problem

The model addresses a reference game. A scene contains several objects that vary in size, colour, and form, and one of them is the intended referent — in our materials, the largest object that matches the target colour and form. A speaker must choose an adjective string that lets a listener identify this referent, and the listener interprets that string by progressively narrowing the set of objects it could pick out. Because objects differ along three properties, the same referent can be described in many ways, and the order in which adjectives are produced is itself part of what the speaker chooses.

## 3.2 Two Mechanisms for Order Sensitivity

Determinate properties such as colour and form are treated as nearly categorical, whereas size is gradable. Whether an object counts as "big" depends on the comparison class it is evaluated against, so the size term is informative only relative to the objects currently under consideration. Against this background, the framework isolates two ways in which adjective order can come to matter. The first is the *speaker architecture*. A global speaker evaluates complete adjective strings and chooses among them as wholes, whereas an incremental speaker chooses one adjective at a time, weighing how the next word would reshape the listener's beliefs at each step. For an incremental speaker, a modifier is evaluated at the moment it is produced, according to the belief update it would cause before later words are heard. The second is the *semantic regime*. Under context-fixed semantics, the size threshold is computed once from the scene and held constant, so reordering the adjectives cannot change what "big" means. Under context-updating semantics, the threshold is recomputed from the listener's running beliefs. Once "blue" has restricted attention to blue objects, "big" is judged against just those objects; "big blue" and "blue big" can then differ even though they contain the same words. This non-commutativity connects to Teodorescu's (2006) observation that ordering restrictions weaken precisely when the alternative orders are not truth-conditionally equivalent.

Crossing these two binary distinctions yields a $2 \times 2$ space of models — speaker architecture (global vs. incremental) by semantic regime (context-fixed vs. context-updating). The global context-fixed model is the baseline and the closest to the vanilla RSA model: meanings are fixed and the speaker evaluates complete strings, so any order preference comes from the shared order prior. The global context-updating model adds posterior-dependent meanings, the incremental context-fixed model adds word-by-word speaker choice, and the incremental context-updating model combines both sources. All four models include the same basic ingredients needed to connect the model to the experimental task: a prior over adjective orders and a utility term based on referential informativeness. The production models additionally include visual salience and stopping components because the production task allows one-, two-, and three-adjective strings; these components are held constant across models so that the $2 \times 2$ comparison isolates the contribution of semantic regime and speaker architecture.

### 3.3 Qualitative Predictions

This framework yields qualitative predictions that the two experiments are designed to test (Table 1). A stable, subjectivity-aligned ordering prior yields a baseline size-first preference that survives even when colour alone identifies the referent (P1). The pragmatic component is sensitive to which property discriminates the referent in the current scene, so the size-first preference should weaken as colour becomes the more useful property (P2). The size-discriminability manipulation provides a proxy-based diagnostic extension of this logic: if lower visual reliability behaves like a display-level analogue of subjectivity, then low size-discriminability contrasts should strengthen the size-first preference (P3). This prediction extrapolates from the compositional logic in Franke et al. (2019); Scontras et al. (2020): if lower size discriminability makes size interpretation noisier, placing colour closer to the noun restricts the comparison class before size is evaluated, which corresponds to a size-first order preference in the German prenominal sequence. Finally, because production allows the full inventory of one-, two-, and three-adjective strings, the framework predicts a graded over-informativity gradient that the binary slider task cannot reveal; if low size discriminability makes size less reliable for reference, displays in the low size-discriminability condition should elicit longer, more redundant strings (P4).

**Table 1**

*Qualitative predictions of the modelling framework and where each is tested.*

| Label | Prediction | Tested in |
|---|---|---|
| P1 | A baseline size-first pressure is detectable even when colour alone identifies the referent. | Exp. 1 & 2 |
| P2 | The size-first preference weakens as colour becomes the more discriminating property. | Exp. 1 & 2 |
| P3 | If low visual size discriminability behaves like a display-level proxy for subjectivity, it yields a stronger size-first preference. | Exp. 1 & 2 |
| P4 | The production task exposes a graded over-informativity gradient across one-, two-, and three-adjective strings, and that gradient is modulated by size discriminability. | Exp. 2 |

# 4 Experiments

We tested the qualitative predictions of the modelling framework (Table 1) with two complementary online referential tasks in German. Experiment 1 measured explicit order preferences in a slider-rating task, whereas Experiment 2 measured adjective selection and ordering in constrained production. Comparing the two tasks lets us ask whether the same context manipulations affect both direct judgments between competing orders and speakers' own adjective choices.

## 4.1 Shared Method

### *4.1.1 Design*

Both experiments shared the same basic referential design. The response format differed across experiments, but the visual displays and target-definition logic were shared. On every critical trial, participants saw a six-object display with one framed target. The target was always the largest object that matched the intended colour and form. The analyses reported here focus on the theoretically central size–colour subset.[1] Within that subset, referential context varied within participants and had three levels: size-sufficient, colour-sufficient, and both-necessary. A second, between-participant factor, size discriminability, varied the size gap between the target and its size competitors. High size-discriminability displays used larger target–competitor gaps, making the size contrast visually pronounced, whereas low size-discriminability displays used smaller gaps, making size a less reliable cue. This factor was manipulated between participants so that each participant experienced a consistent level of visual size contrast throughout the task (Table 2).

**Table 2**

*Shared design of the two experiments.*

| **Factor** | **Levels** |
| --- | --- |
| Referential context | Size-sufficient, colour-sufficient, both-necessary |
| Size discriminability | High (strong size contrast), low (weak size contrast) |

#### *4.1.2 Materials*

An example item is shown in Figure 2. Items were distributed across participants in a Latin-square design, so that scene templates and condition assignments were counterbalanced across lists. In addition to list counterbalancing, the critical scenes were realised through systematic permutations of colour and form values, creating variability in the surface displays while preserving the referential structure required by each condition. These permutations were applied to 27 critical scene templates, each specifying an abstract six-object configuration with a target, competitors, and distractors. Critical vocabularies

[1] The broader paradigm also included size–form and colour–form trials to widen the material base and reduce dependence on a single adjective pairing. These conditions remain part of the experimental design, but the present analysis concentrates on the size–colour subset because it provides the clearest contrast between a gradable adjective and a determinate feature adjective. The exact size values, target–competitor gap construction, and list assignment are reported in Appendix A.

were fixed across the study, with six colour terms and six form terms reused throughout the scene set; the modified noun was always *Aufkleber* ('sticker').[2] For the computational model, each display was represented as a $6 \times 3$ matrix, with one row per object and three columns for size ($s$), colour ($c$), and form ($f$). The referent was stored at index 0, but participants saw the displays as ordinary visual arrays rather than as explicit feature lists.

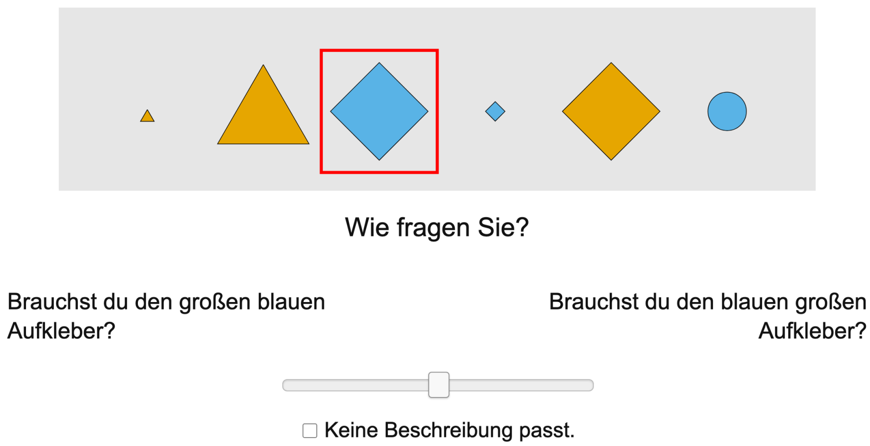



**Figure 2**

*Example slider-rating trial from the experiment.*

#### *4.1.3 Procedure*

Both experiments were conducted online via Prolific. Participants in each experiment were instructed to identify the framed target using its perceptual properties, not spatial descriptions such as *left* or *right*. The task was framed as a simple reference game about collectible stickers: participants saw one framed target and gave or evaluated a description for a partner who could see the same stickers in a different order but could not see the frame. This made spatial terms uninformative and kept the task focused on adjective descriptions. Both experiments began with a consent form and a brief practice block (three training trials

[2] The form values were presented with German adjectival forms, such as *runden* for round/circle and *quadratischen* for square, so that form could occupy the same prenominal adjective position as colour and size.

with feedback) that familiarised participants with the interface and the referential task. The main trials were divided into four blocks of approximately equal length, with self-paced breaks between blocks. Trial order within each block was randomised independently for each participant. The experiment-specific response formats, trial counts, and response coding are reported below.

## 4.2 Experiment 1: Slider-Rating Study

### *4.2.1 Participants*

Participants ($N = 134$ after exclusions; 70 in the LOW size discriminability group, 64 in the HIGH size discriminability group) were recruited online via Prolific and reported German as their native language. Exclusion criteria were: more than three missing trial-list entries, completion times more than two standard deviations from the sample mean, more than three failures on control items, or more than three "none fits" responses on experimental items.[3] After exclusions, the analysed dataset contained 10,606 critical observations. The size–colour subset reported in the main empirical and model-comparison figures contained 3,485 observations; exclusion details are summarized in Appendix A. Mean completion time was 27.6 minutes. Participants were compensated via Prolific at a rate above the platform-recommended minimum.

### *4.2.2 Procedure*

Experiment 1 implemented the shared referential design as a comparative judgment task. On each critical trial, two competing adjective orders appeared on opposite sides of a continuous slider, together with a "none of the descriptions fits" option. Left–right presentation of the two orders was randomised across trials. Each participant saw one of six lists built from the 27 underlying scenes, yielding 81 critical trials and 99 filler and control

[3] Control items were filler trials with an unambiguous expected response, included to screen for inattentive or non-referential responding. The additional "none fits" option allowed participants to indicate that neither description identified the target; frequent use of this option on experimental items was treated as evidence that the participant was not performing the intended comparative-judgment task.

trials (180 total). The filler and control items discouraged fixed response strategies and ensured attention to the referential task. Responses were recorded on a 0–100 scale and centred for analysis so that positive values indicated a size-first preference.

#### *4.2.3 Results*

Figure 3 shows that size-first orderings were preferred across all three referential contexts, including colour-sufficient displays where size did not identify the target by itself, but the strength of this preference varied across contexts and size-discriminability levels. We analysed centred slider ratings with linear mixed-effects models fit in R using `lme4::lmer` (Bates, Mächler, Bolker, & Walker, 2014). The full model included referential context, size discriminability, and their interaction as fixed effects, with random intercepts for participant and item; effects were evaluated by likelihood-ratio comparisons between nested models.

The interaction between referential context and size discriminability was significant, $\chi^2(2) = 11.7$, $p = .003$. Breaking down this interaction, higher size discriminability slightly increased the size-first preference in size-sufficient contexts (low: 0.30, high: 0.34) but slightly reduced it in both-necessary (0.25 vs. 0.20) and colour-sufficient contexts (0.06 vs. 0.04). However, Bonferroni-corrected pairwise comparisons showed that none of the high–low size-discriminability contrasts was significant within an individual referential context (all $p \geq .23$). We therefore interpret the interaction as a weak crossover. In size-sufficient displays, high size discriminability slightly strengthened the already strong size-first preference, whereas in colour-sufficient displays it slightly reduced size-first ratings, matching the direction made salient by referential context in each case. This pattern suggests that high size discriminability may amplify the cue already supported by the referential context. In both-necessary displays, the contrast went in the direction predicted by P3, with lower size discriminability associated with a stronger size-first preference, but this difference was not significant after correction.

Further analysis revealed that referential context had a significant main effect, $\chi^2(2) = 402.6$, $p < .001$. Mean centred ratings, indexing preference for size-first orderings, were highest in size-sufficient displays, lower in both-necessary displays, and lowest in colour-sufficient displays. Even colour-sufficient contexts did not produce a qualitative reversal: mean centred ratings remained above zero.

Together, these patterns indicate two effects in the slider data. First, participants retained a size-first bias even when colour was sufficient for identifying the target, supporting P1. Second, this bias was graded by referential usefulness: it was strongest when size alone identified the target and weakest when colour alone did, supporting P2. The display-level proxy prediction for P3 receives limited support. The both-necessary contrast points in the predicted direction, but the contrast is not significant, and the one-property-sufficient contexts suggest weak salience-linked amplification of the contextually useful property. Size discriminability therefore showed only a small context-linked modulation of size-first ratings, not a robust monotonic effect of lower visual size reliability.

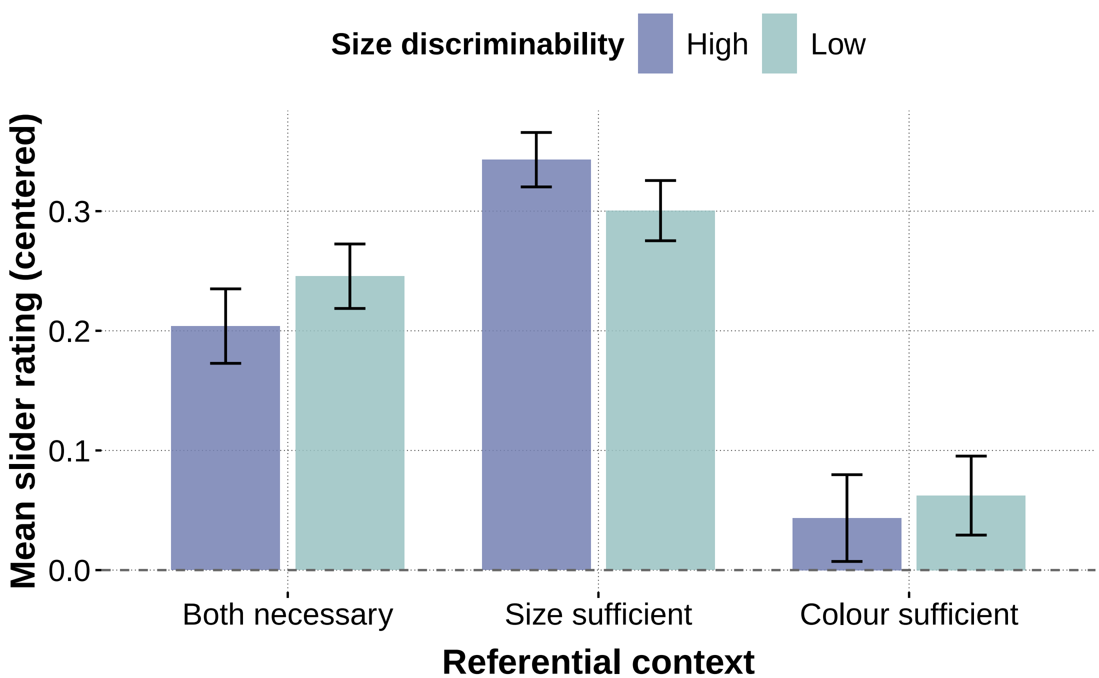



**Figure 3**

*Slider-rating data. Mean centred slider ratings by referential context and size discriminability, with 95% confidence intervals; positive values indicate size-first preference.*

## 4.3 Experiment 2: Constrained-Production Study

### *4.3.1 Participants*

Participants ($N = 113$ after exclusions; 60 in the LOW size discriminability group, 53 in the HIGH size discriminability group) were recruited online via Prolific; anyone who had completed the slider study was excluded. Further exclusion criteria were: completion times more than two standard deviations from the mean, more than 80% identical responses, or more than 10% missing or unannotatable responses. The full preprocessed data contained 9,639 responses across the three adjective-pair families. The size–colour production model reported in Section 6.2 used 3,196 annotated observations; exclusion and coding details are summarized in Appendix A. Mean completion time was 32.4 minutes. Compensation followed the same Prolific rate as Experiment 1.

### *4.3.2 Procedure*

Experiment 2 used the shared referential design in a constrained production format. Participants saw the framed target together with the fixed sentence frame *den* _____ *Aufkleber* and typed the adjective material needed before the noun *Aufkleber*. Restricting the response to the adjective slot rather than eliciting a full sentence reduced off-task variability and provided a controlled basis for coding. The practice materials additionally introduced the relevant colour and form vocabulary before the critical trials began. Each participant completed 135 main trials: 81 critical trials and 54 filler or control trials from one of six counterbalanced lists.[4]

Responses were coded into 15 utterance categories corresponding to all ordered combinations of S (size), C (colour), and F (form): 3 one-adjective, 6 two-adjective, and 6 three-adjective strings. Misspellings and transparent lexical alternatives were normalised when their semantic class was unambiguous; spatial descriptions, comparative or ordinal size expressions, and other unclassifiable responses were excluded. The dependent variable for the

[4] The number of filler and control trials was reduced in Experiment 2 to keep the total number of trials manageable because typed production responses were slower and more effortful than slider judgments.

Bayesian model comparison is the categorical utterance label $y_i \in \{1, \ldots, 15\}$ for each trial $i$. For descriptive statistics and targeted mixed-effects analyses, we also tracked two derived binary outcomes: whether an utterance was size-initial and whether it was over-informative (i.e., a three-adjective response in both-necessary contexts, or a multi-adjective response in size-sufficient or colour-sufficient contexts). For the mixed-effects logistic regressions, trials satisfying the coded outcome were coded as 1 and all other annotated trials were coded as 0.

#### *4.3.3 Results*

Figure 4 shows that constrained production separates two patterns that the slider task could not distinguish: which adjective material speakers selected and, when more than one adjective was produced, which adjective came first. Size-sufficient displays mostly elicited size-initial descriptions, whereas colour-sufficient displays mostly elicited colour-only descriptions, leaving little multi-adjective material on which a colour-first ordering preference could be expressed. Both-necessary displays fell between these extremes and elicited more multi-adjective strings. This response structure motivates two complementary analyses: a size-initial model for first-adjective choice and an over-informativeness model for whether speakers selected redundant adjective material. We analysed two derived binary outcomes with mixed-effects logistic regressions fit in R using `lme4::glmer` (Bates et al., 2014). The first model coded whether the response was size-initial; the second coded whether the response was over-informative. Both models included referential context, size discriminability, and their interaction as fixed effects, with random intercepts for participant and item; effects were evaluated by likelihood-ratio comparisons between nested models.

For size-initial responses, the interaction between referential context and size discriminability was significant, $\chi^2(2) = 31.4$, $p < .001$. Bonferroni-corrected simple contrasts within each referential context showed significant high–low size-discriminability differences in all three contexts (all $p < .001$). Higher size discriminability slightly reduced size-initial responses in size-sufficient contexts (low: 0.80, high: 0.77) and colour-sufficient contexts (0.04 vs. 0.01), but increased size-initial responses in both-necessary contexts (0.49

vs. 0.65). This pattern differs from the slider interaction. In production, sharper size contrasts promoted size-initial responses only when both size and colour were needed; when one property was sufficient, speakers could often resolve the task by selecting less adjective material, so size discriminability affected adjective selection and utterance length in addition to ordering. This motivates the over-informativity analysis below. Further analysis revealed a strong main effect of referential context, $\chi^2(2) = 1859.7$, $p < .001$, but no significant main effect of size discriminability, $\chi^2(1) = 0.79$, $p = .37$. Size-initial responses were most frequent when size alone identified the target, least frequent when colour alone did, and intermediate when both properties were needed. This follows the same contextual ordering as the slider data, but the production responses were more categorical: colour-sufficient displays mostly elicited colour-only descriptions, not weak size-first preferences.

A complementary analysis of over-informativity tested the adjective-selection side of the production data. The interaction between referential context and size discriminability was significant, $\chi^2(2) = 63.8$, $p < .001$. The interaction was driven mainly by size-sufficient displays: low size-discriminability displays elicited more over-informative responses than high size-discriminability displays (low: 0.90, high: 0.63). Bonferroni-corrected simple contrasts showed that this low–high difference was significant in size-sufficient contexts ($p < .001$). The corresponding descriptive contrasts were smaller in both-necessary contexts (0.40 vs. 0.38) and colour-sufficient contexts (0.33 vs. 0.26). Further analysis revealed a strong main effect of referential context, $\chi^2(2) = 725.7$, $p < .001$, and a main effect of size discriminability, $\chi^2(1) = 10.3$, $p = .001$.

Together, these patterns indicate four effects in the production data. First, adjective selection and ordering were strongly graded by referential usefulness: size-initial responses increased when size was useful and decreased when colour was sufficient, supporting P2. Second, P1 receives a task-qualified interpretation in production: colour-sufficient displays did not mainly reveal a competing colour-first order over size, because speakers usually selected a single colour adjective. The production task therefore shows how adjective selection can bypass the size-first baseline, which is why over-informativeness is informative in this response format. Third, the display-level proxy prediction for P3 is not supported. The size-initial interaction is asymmetric, and lower visual size reliability does not monotonically promote size-first ordering. Fourth, P4 is supported: the richer production response space reveals a graded over-informativity pattern across one-, two-, and three-adjective strings, consistent with work showing that redundant modifiers can support efficient reference when facing increased uncertainty about contexts (Degen et al., 2020; Rubio-Fernández, 2016; Rubio-Fernandez, 2019). In the present data, low size-discriminability displays elicited more redundant strings overall.

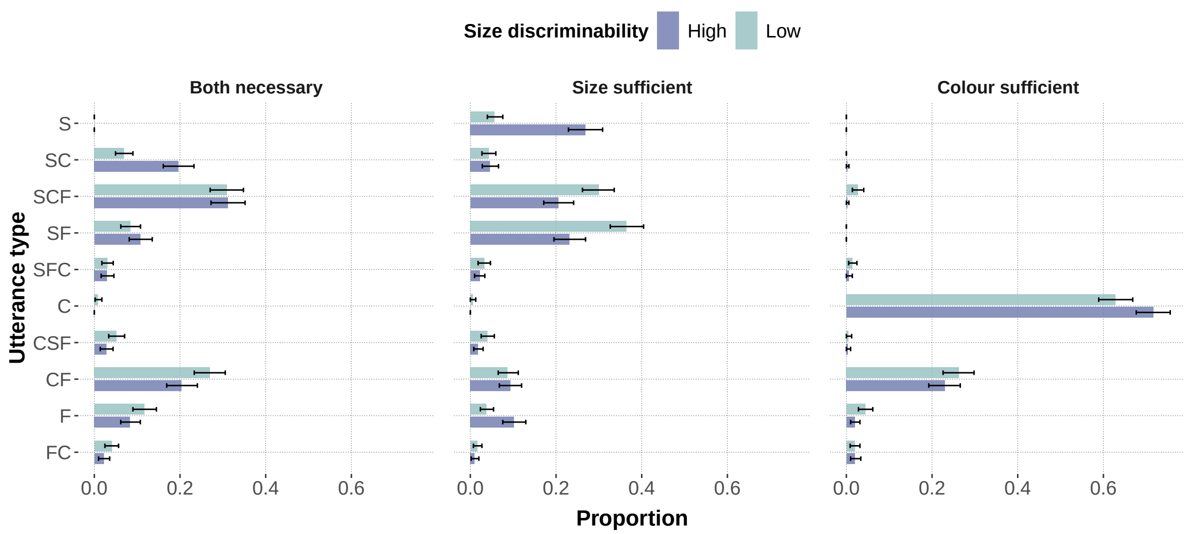



**Figure 4**

*Constrained-production data. Utterance-type proportions by referential context and size discriminability, with bootstrapped 95% confidence intervals; only utterance types reaching at least 2% in one condition are shown.*

## 4.4 Interim Summary

Experiment 1 shows that adjective-order judgments retain a size-first baseline even when size is not necessary for reference. That baseline is context-sensitive: ratings were strongest when size identified the target and weakest when colour did. Size discriminability did not add a robust simple effect; it produced a weak crossover without significant high–low contrasts within individual contexts.

Experiment 2 extends this pattern to constrained production, where participants selected the adjective directly for the target description. Speakers tended to use size-initial descriptions when size was useful, but colour-sufficient displays often elicited a single colour adjective. Reversed colour-before-size orders were rarely produced; when colour alone identified the target, speakers more often resolved the task through adjective selection before an ordering contrast between size and colour arose. The production data also exposed an utterance-length effect: lower size discriminability elicited more redundant adjectives, especially in size-sufficient displays.

The two tasks locate context sensitivity at two related levels. Slider judgments show how referential context grades an explicit ordering preference. Production shows that the same referential context can first determine which adjectives are selected, and only then which order is used. This link helps interpret the apparent mismatch in the size-discriminability interactions: visual size reliability did not monotonically shift responses toward size-first ordering, but it did affect whether speakers added extra adjectives.

The formal modelling task is to account for these patterns within a shared representational framework. A successful model should preserve the size-first baseline, allow adjective choice and ordering to vary with referential usefulness, and represent the richer production response space over one-, two-, and three-adjective strings.

## 5 Computational Model

This section turns the modelling framework introduced in Section 3 into a formal RSA model. The formalization separates components shared across model variants from components that instantiate the two tested contrasts: speaker architecture and semantic regime. The section specifies the representational primitives, literal-listener semantics, global and incremental speakers, context-fixed and context-updating regimes, and inference setup for the $2 \times 2$ model comparison. The experiments were broader than the fitted comparison:

the full paradigm included size–form and colour–form trials, but the analyses reported here focus on size–colour scenes, where the contrast between a gradable adjective and a determinate feature adjective is clearest. The fitted analyses therefore use colour as the determinate feature in the main gradable–determinate contrast.

## 5.1 Representational Primitives

An object is a triple $o = (s, c, f)$, where $s \in [1, 10]$ denotes size, $c \in \{0, 1\}$ indicates colour match, and $f \in \{0, 1\}$ indicates form match. A context $\mathcal{C}$ is a set of $n_{\text{obj}}$ objects. In the experiments, the intended referent is always the largest object with $c = f = 1$. The size–colour fits use C as the determinate feature that competes with size for identification. The formal vocabulary still includes F because form was available in the broader experiment and could appear as an additional, potentially redundant modifier in production.

Utterances are built from three adjective types, S, C, and F. An utterance $u = (a_1, \ldots, a_T)$ is an ordered sequence of adjectives $a_t \in \{\mathrm{S}, \mathrm{C}, \mathrm{F}\}$ with $T \in \{1, 2, 3\}$. Combining all lengths and orderings yields 15 surface forms: {S, SC, SCF, SF, SFC, C, CS, CSF, CF, CFS, F, FS, FSC, FC, FCS}. This full inventory reflects the broader experimental vocabulary retained by the production model.

## 5.2 Meaning Functions

**Context-dependent size semantics.** We model size adjectives with a context-sensitive threshold semantics following Schmidt, Goodman, Barner, and Tenenbaum (2009), who showed that the interpretation of gradable adjectives depends on the statistics of the comparison class. Franke et al. (2019) adopted the same approach in their simulation of adjective-ordering preferences. A threshold is computed from the current comparison class $\mathcal{C}$:

$$\theta_k(\mathcal{C}) = x_{\max}(\mathcal{C}) - k\left(x_{\max}(\mathcal{C}) - x_{\min}(\mathcal{C})\right), \tag{1}$$

where $x_{\min}(\mathcal{C})$ and $x_{\max}(\mathcal{C})$ are the lower and upper bounds of the mid-range of the contextual size distribution, and $k \in [0, 1]$ controls the strictness of the threshold. Smaller values set a higher, more selective standard for “big”; larger values make the adjective apply more broadly. The resulting semantic value $[\![\mathrm{S}]\!](o, \mathcal{C}) \in [0, 1]$ is graded rather than binary: an object's size is compared to the threshold via a normal slack function scaled by a perceptual-blur parameter $w_f$, consistent with scalar variability in the Approximate Number System (Feigenson, Dehaene, & Spelke, 2004). This context dependence is what makes the context-fixed vs. context-updating distinction possible: because $\theta_k$ depends on $\mathcal{C}$, changing which objects are in the comparison class changes what counts as big.

**Colour and form semantics.** Colour and form are in principle determinate: an object either matches the feature or it does not. We follow the relaxed-semantics approach of Degen et al. (2020) and Waldon and Degen (2021) by softening these denotations with $\nu \in (0, 1)$, which can be interpreted as the probability that a feature-matching object is correctly described by the adjective. This relaxation allows even determinate properties to carry graded information, which is needed for modelling redundant modification. In the reported model fits, colour is the primary determinate comparison case because inference is carried out on size–colour scenes, whereas form is retained as a general determinate feature term and as a possible redundant modifier in production. In the slider models, colour and form share a single $\nu$ parameter. In the production models, they receive separate fixed values, $\nu_{\text{colour}}$ and $\nu_{\text{form}}$, whose provenance and sensitivity checks are summarized in Appendix D; the values are held constant across all final model variants.

## 5.3 Incremental Listener

The literal listener constructs utterance meanings incrementally by combining token-level semantic contributions while updating the effective comparison class. Let $u = (w_1, \ldots, w_T)$ be an utterance. The listener maintains a sequence of belief states $\{P_t(o)\}_{t=0}^{T}$, where $P_t(o)$ denotes the posterior belief after incorporating the suffix $(w_{t+1}, \ldots, w_T)$:

$$P_t(o) = P\Big(o \mid w_{t+1}, \ldots, w_T\Big), \qquad t = 0, \ldots, T, \tag{2}$$

with base case

$$P_T(o) = P_{\text{prior}}(o). \tag{3}$$

For $t = T, T-1, \ldots, 1$, beliefs are updated according to

$$P_{t-1}(o) \propto P_t(o)\, [\![w_t]\!](o), \tag{4}$$

which normalizes to

$$P_{t-1}(o) = \frac{P_t(o)\, [\![w_t]\!](o)}{\sum_{o'} P_t(o')\, [\![w_t]\!](o')}. \tag{5}$$

After all tokens have been processed, the listener arrives at

$$P_0(o) = P(o \mid u). \tag{6}$$

The reverse indexing reflects prenominal composition: material closer to the noun establishes the comparison class for material farther from the noun. The composition rule in Equation (4) is shared by both semantic regimes of the model. What differs is how the meaning function $[\![w_t]\!]$ is computed for gradable adjectives such as size.

**Context-updating semantics.** Under the context-updating regime, the size threshold $\theta_k$ is recomputed from the listener's current belief state at each composition step:

$$[\![w_t]\!]_{\text{UPD}}(o) \;=\; [\![w_t]\!]_{P_t}(o), \tag{7}$$

where $P_t$ is the posterior after processing all words to the right of position $t$. The current belief state therefore plays a dual role: it serves as the prior for the next update step, and it determines the semantics of the next token. Because $P_t$ itself depends on the semantics of later words, this creates a feedback loop:

$$[\![w_t]\!]_{P_t} \quad \text{depends on} \quad P_t(o) \propto P_{t+1}(o)\, [\![w_{t+1}]\!]_{P_{t+1}}(o). \tag{8}$$

Word order changes the sequence of comparison classes and thereby the interpretation of the utterance, yielding non-commutativity:

$$P(o \mid w_1, w_2) \neq P(o \mid w_2, w_1) \quad \text{in general.} \tag{9}$$

**Context-fixed semantics.** Under the context-fixed regime, the size threshold is computed once from the scene prior and held constant throughout composition:

$$[\![w_t]\!]_{\text{FIX}}(o) \;=\; [\![w_t]\!]_{P_{\text{prior}}}(o) \quad \text{for all } t. \tag{10}$$

The listener's beliefs still update incrementally via Equation (4), but this update no longer feeds back into the meaning function. For deterministic adjectives such as colour and form, both regimes are identical because their semantics does not depend on the comparison class. The context-fixed regime therefore isolates the contribution of incremental belief updating from the additional effect of posterior-dependent meaning. The fitted models below test this specific form of posterior-dependent thresholding. They leave open stronger formulations of context dependence, such as treating a preceding colour adjective as a hard restriction on the comparison class or using a narrower size-membership function. This point matters for interpreting the semantic-regime comparison: if the colour update only weakly changes the size threshold, context-updating semantics has little room to improve predictive fit.

### 5.4 Utterance Prior and Speaker Models

The model incorporates a prior over utterance forms that is separate from the pragmatic component. This prior captures stable ordering tendencies — such as the well-documented subjectivity-driven size-first preference — that the speaker brings to the task before any scene-specific pragmatic reasoning.

In the slider-rating models, the prior is minimal: a single inferred bias parameter $b \geq 0$ that penalizes the reverse (colour-first) order relative to the canonical (size-first) order. Because $b$ is estimated from data rather than fixed, the posterior indicates how much residual ordering preference remains after the RSA mechanism has been applied; if the pragmatic model were sufficient on its own, the posterior for $b$ would concentrate near zero.

In the production models, the prior is derived from a language-model distribution over the 15 utterance forms, estimated from German GPT-2 surprisal values averaged across the experimental adjective–noun combinations. To keep this prior from explaining away utterance length or adjective content, we use only its residual order preference within each unordered content set. For example, the prior can distinguish SC from CS, but it cannot by itself prefer two-adjective over one-adjective descriptions, or colour descriptions over size descriptions. The contribution of this order-only prior is scaled by an inferred parameter $\beta$. Like $b$ in the slider model, it encodes stable word-order tendencies from language experience, which the pragmatic speaker then modulates in light of the current scene. Against this shared listener and prior, the two speaker architectures differ in when pragmatic utility is evaluated.

**Global speaker.** The global speaker computes a distribution over complete utterances by applying a softmax to utterance-level utility:

$$S_{\text{gb}}(u \mid o) = \frac{\exp\Big(U(o,u)\Big)}{\sum_{u'} \exp\Big(U(o,u')\Big)}. \tag{11}$$

For the shared production core, utility is

$$U(o^*, u) = \alpha \, \log L(o^* \mid u) - \beta \, \Omega(u), \tag{12}$$

where $\alpha$ is the RSA rationality parameter and $\Omega(u)$ is the order-only language-model score. The production model adds the salience, stopping, and lapse terms described in Section 6.2 to this core utility. In the slider-rating models, the utility simplifies to

$$U(o,u) = \alpha\Big(\log L(o \mid u) - b_u\Big), \tag{13}$$

where $b_u = 0$ for the size-first order and $b_u = b$ for the reverse order.

**Incremental speaker.** The incremental speaker chooses one token at a time. For an intended referent $o^*$,

$$S_{\text{inc}}(u \mid o^*) = \prod_{t=1}^{T} S_t(w_t \mid u_{<t}, o^*)\,, \qquad u_{<t} = (w_1, \ldots, w_{t-1}), \tag{14}$$

with local decision rule

$$S_t(a \mid u_{<t}, o^*) \propto \Big[L(o^* \mid u_{<t} + a)\Big]^{\alpha}. \tag{15}$$

Combining local pragmatic choice with the order-only prior yields the shared incremental production core:

$$S_{\text{inc}}(u \mid o^*) \propto \exp\{\beta\,\Omega(u)\} \cdot \prod_{t=1}^{T} \frac{\Big[L(o^* \mid u_{<t} + w_t)\Big]^{\alpha}}{Z_t(u_{<t}, o^*)}, \tag{16}$$

where

$$Z_t(u_{<t}, o^*) = \sum_a \Big[L(o^* \mid u_{<t} + a)\Big]^{\alpha} \tag{17}$$

ensures local normalization. In the slider-rating models, the LM prior is replaced by the bias term: $S_{\text{inc}}(u \mid o^*) \propto \exp(-b_u) \cdot \prod_t S_t(w_t \mid u_{<t}, o^*)$. The theoretical contrast is that the global speaker applies pragmatic reasoning once over complete utterances, whereas the incremental speaker applies it at each token position. When semantics is context-dependent, these two model variants are not equivalent. The equations in this subsection define the shared pragmatic core. The constrained-production fits then apply the same production extensions — order-only prior, visual-salience terms, salience-sensitive stopping cost, and lapse mixture — in every model of the production $2 \times 2$.

## 6 Bayesian Model Comparison

The $2 \times 2$ comparison crosses the two mechanisms introduced in Section 3.2: speaker architecture (global vs. incremental) and semantic regime (context-fixed vs. context-updating). The comparison is fitted independently to both datasets with hierarchical participant pooling. Within each dataset, the four models use matched parameter inventories. Fit differences are therefore interpreted as evidence about speaker architecture and semantic regime rather than as a consequence of unequal expressive flexibility, subject to the diagnostic caveats reported below and in the appendix. The comparison is diagnostic in three ways. A consistent advantage for incremental over global speakers would indicate that word-by-word choice contributes predictive accuracy beyond utterance-level choice. A consistent advantage for context-updating over context-fixed semantics would indicate that recomputing gradable meanings from the compositionally updated comparison class contributes predictive accuracy beyond incremental belief updating alone. A positive difference-in-differences would indicate that the value of context-updating semantics depends on whether production itself is incremental. We use Pareto-smoothed importance-sampling leave-one-out cross-validation (PSIS-LOO; Vehtari et al., 2017). For each comparison, we report differences in expected log pointwise predictive density ($\Delta$ELPD) relative to the best-fitting model together with the difference standard error (dse). A ratio $|\Delta\text{ELPD}|/\text{dse}$ substantially greater than 1 is treated as evidence for a clear difference in out-of-sample predictive accuracy. All models were implemented in NumPyro (Phan, Pradhan, & Jankowiak, 2019) and estimated with four MCMC chains using the NUTS sampler (Hoffman & Gelman, 2014). All models reported in the main text use hierarchical participant-level pooling, implemented as participant-level offsets on the rationality parameter. The incremental production models that drive the main comparison converged cleanly; diagnostics for the global robustness check are reported in Appendix C. The inferential focus is the comparison among incremental models, because these models both fit the data well and have acceptable MCMC diagnostics. Global models are retained as structurally informative baselines, but when they combine very poor predictive fit with weak diagnostics, their parameter estimates should not be interpreted substantively. Appendix C reports the available PSIS-LOO warnings, convergence summaries, and production Pareto-$k$ counts for the main comparisons. Dataset-specific parameter inventories, likelihoods, and sampler settings are introduced with each subsection below.

### 6.1 Slider-Rating Study

The slider $2 \times 2$ is fitted on a minimal speaker with a single rationality parameter $\alpha$, a bias intercept $b$, and a participant-level offset, with the semantic constants held fixed at $k = 0.5$, $w_f = 1.0$, and $\nu = 0.80$. Slider responses cluster at the boundaries (0 and 1) and are modelled with a Zero-One-Inflated Beta likelihood; each chain drew 500 warmup and 500 post-warmup samples (target acceptance probability = 0.90, maximum tree depth = 8).

Figure 5 shows the $2 \times 2$ comparison for the slider data. The largest contrast is speaker architecture: both incremental models fit the data well and are statistically

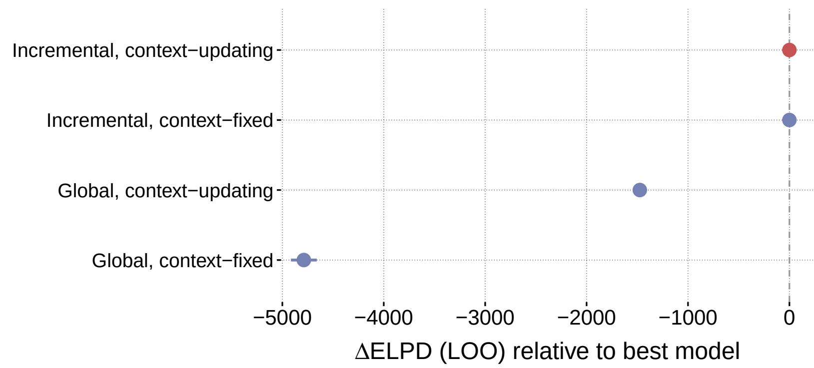
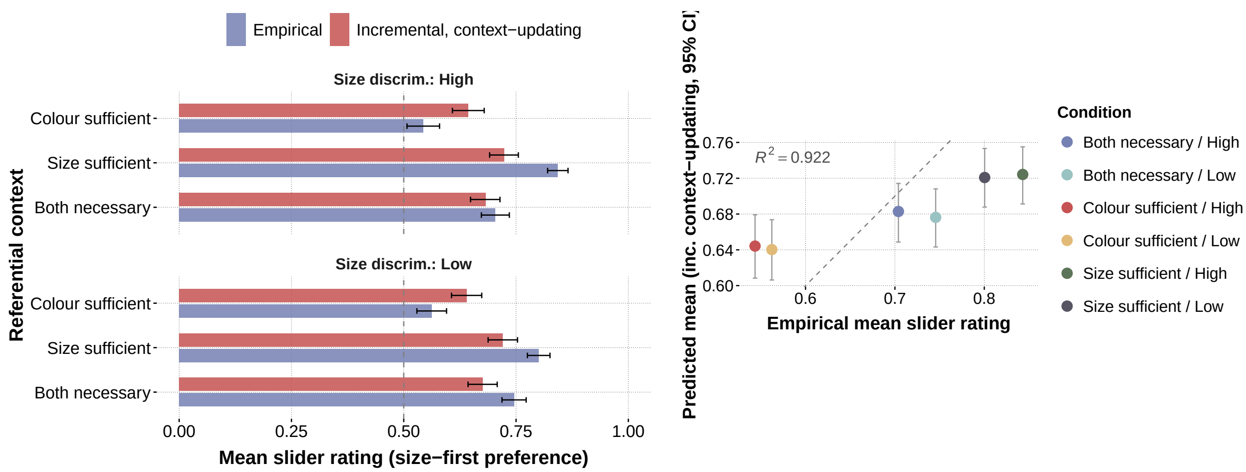



**Figure 5**

*Slider-rating model comparison: $2 \times 2$ speaker $\times$ semantics. $\Delta$ELPD relative to the best model, with $\pm 1$ difference standard error.*




(a) Mean rating by condition

(b) Condition-level correlation

**Figure 6**

*Posterior-predictive fit of the incremental, context-updating slider model. (a) Observed mean slider ratings and posterior-predictive model means by referential context $\times$ size discriminability; dashed line marks* 0.5 *(no preference), error bars are* 95% *confidence and posterior-predictive intervals. (b) Observed vs. predicted condition means; dashed line is identity.*

indistinguishable ($\Delta\text{ELPD} = 0.06$, $\text{dse} = 0.18$, $|\Delta\text{ELPD}|/\text{dse} = 0.32$), whereas both global models perform far worse ($\Delta\text{ELPD} > 1{,}400$). Within the incremental pair, context-updating semantics confers no detectable advantage over context-fixed semantics. For graded preference judgments between two competing orders, the predictive gain is carried by incremental pragmatic evaluation. This result does not show that posterior-dependent size meanings are absent; it shows that, within the incremental slider models, recomputing the size threshold from the running posterior does not improve out-of-sample prediction. Posterior-predictive checks show that the incremental context-updating model captures the condition-level ordering well, tracking the empirical ordering across all six referential context $\times$ size-discriminability combinations with $r = 0.96$ (squared Pearson $\approx 0.92$; Figure 6). The model slightly under-predicts the strongest size-first conditions (size-sufficient) and slightly over-predicts the weakest (colour-sufficient), compressing the predicted range relative to the empirical range while preserving the rank-ordering. For the best-fitting incremental context-updating model, the posterior rationality parameter was $\alpha = 0.69$ ($SD = 0.06$, 94% HDI $[0.57, 0.79]$). The implied Beta concentration of the continuous ZOIB component was $\varphi = 1/\sigma^2 \approx 1.93$ (94% HDI $[1.85, 2.04]$), corresponding to Beta$(\mu \cdot 1.93, (1 - \mu) \cdot 1.93)$. This value implies a broad, near-uniform spread around the conditional mean, placing most of the conditional structure in $\mu$ and in the boundary inflations. The hierarchical spread of participant-level rationality offsets was $\tau = 0.12$ ($SD = 0.01$, 94% HDI $[0.10, 0.14]$). The residual bias intercept was $b = 0.58$ ($SD = 0.06$, 94% HDI $[0.47, 0.69]$), with posterior probability 0.91 above 0.50. This positive bias means that even after the RSA mechanism accounts for context-dependent informativeness, the model infers a residual size-first preference — consistent with the view that a stable ordering default persists beyond what discriminatory utility alone explains. The binary slider response space does not strongly discriminate between sharper and softer colour predicates: the slider posterior for $\nu$ is broad (0.63, 94% HDI $[0.50, 0.82]$), places substantial mass near the lower prior boundary at 0.5, only just excludes 0.80, and the four models remain statistically indistinguishable in PSIS-LOO across this range (all $|\Delta\text{ELPD}|/\text{dse} < 1$; full numbers in Appendix F). The residual-bias estimate is conditional on the fixed $\nu = 0.80$ specification: under softer colour predicates the rationality $\alpha$ and the bias $b$ shift substantially (towards 1.5 and below 0.5, respectively). Within the global pair, context-updating semantics substantially outperforms context-fixed semantics ($\Delta$ELPD $\approx$ 3,313, $|\Delta\text{ELPD}|/\text{dse} > 24$). Context-updating composition thus provides a minimal source of order sensitivity that partially compensates for the lack of incrementality, but both global models remain far worse than either incremental variant.

### 6.2 Constrained-Production Study

Constrained production is a fifteen-way choice over one-, two-, and three-adjective strings, so the production speaker requires more than the binary bias used for slider ratings. At the same time, the production model must not encode each empirical regularity with a separate condition-specific parameter. We therefore use a parsimonious speaker with six population parameters and three interpretable components. These components instantiate the division of labor introduced in Section 3.2: stable order expectations, non-pragmatic visual prominence, and listener-oriented utility are represented separately and held constant across all four models of the architecture $\times$ semantics comparison. No component directly encodes referential context as a condition label; size discriminability enters through display-derived quantities such as target–competitor size contrast and the sharpness adjustment in the salience term.

**Principled production components.** First, stable ordering preferences enter through the order-only language-model score $\Omega(u)$ introduced above. Because $\Omega(u)$ is residualized within unordered content sets, it contributes a baseline ordering preference without carrying independent information about content choice or utterance length. Second, each adjective receives a soft visual-salience score. The base salience order is colour > form > size, reflecting the salience and consistency of colour in referring-expression production relative to size (Tarenskeen, Broersma, & Geurts, 2015), but the score is adjusted by the current display. Let $\eta = (0, 1, 0.25)$ be the fixed base salience vector for size, colour, and form. For each trial, we compute a context adjustment

$$m = \begin{pmatrix} \sigma\left(\frac{s_0 - \max_{i>0} s_i}{\mathrm{sd}(s)+10^{-8}}\right)(0.5 + 0.5\,\mathbb{I}_{\mathrm{sharp}}) \\ 1 - \frac{1}{n-1}\sum_{i>0}\mathbb{I}[c_i = c_0] \\ 1 - \frac{1}{n-1}\sum_{i>0}\mathbb{I}[f_i = f_0] \end{pmatrix}, \tag{18}$$

where $\sigma$ is the logistic sigmoid, the first term increases with target–competitor size contrast, and the latter two terms increase when fewer distractors share the target colour or form. The trial-specific salience vector is then centered,

$$z = \eta + m - \mathrm{mean}(\eta + m), \tag{19}$$

because only relative salience differences affect the softmax. Salience enters twice. In the incremental speaker it boosts locally salient choices in the token-level decision rule,

$$S_t(a \mid u_{<t}, o^*) \propto \exp\{\alpha \log L(o^* \mid u_{<t} + a) + \lambda_{\mathrm{salience}} z_a\}, \tag{20}$$

For the global production speaker, the same salience scores enter as an utterance-level sum over the adjectives present in the string, so the global models receive the same non-pragmatic salience support without word-by-word choice. In both architectures, salience also imposes a stopping pressure when a salient adjective would be followed by additional material,

$$\mathrm{cost}_{\mathrm{stop}}(u) = \rho_{\mathrm{stop}} \sum_{t<T} \max(z_{a_t}, 0). \tag{21}$$

This captures the tendency for highly salient sufficient colour to appear in short colour-only responses without hard-coding the colour-sufficient condition. Third, a single rationality parameter $\alpha$ controls pragmatic sensitivity to the listener's posterior, and a small lapse rate $\epsilon$ accounts for residual unsystematic choices. The participant hierarchy allows individual differences in pragmatic strength through a shared scale $\tau$.

The determinate feature semantics are held fixed at $\nu_{\mathrm{colour}} = 0.59$ and $\nu_{\mathrm{form}} = 0.50$, with $k = 0.50$ and $w_f = 0.69$ for size. These values are used uniformly across the production comparison. The resulting model has six population parameters: $\alpha$, $\log\beta$, $\lambda_{\mathrm{salience}}$, $\rho_{\mathrm{stop}}$, $\epsilon$, and $\tau$. The uncertainty-based length term considered during model development is not part of the final production model; the final fit is driven by the order-only prior, soft salience, pragmatic informativeness, and salience-sensitive stopping. Appendix D summarizes the model-development checks that led to this inventory.

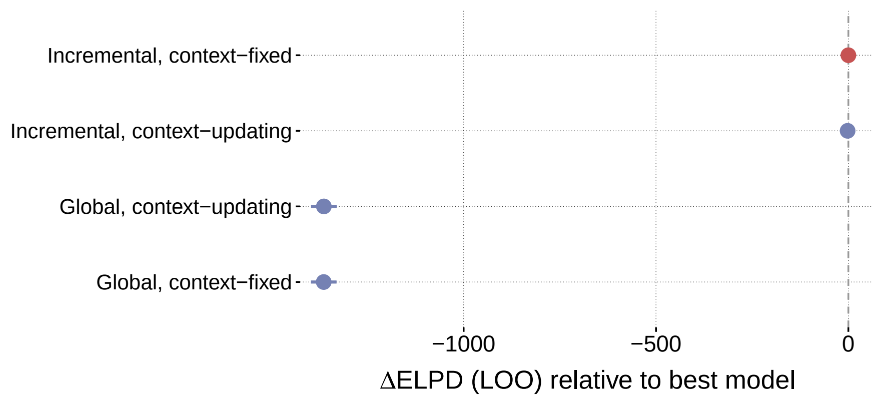



**Figure 7**

*Constrained-production model comparison: PSIS-LOO ΔELPD relative to the best model for the parameter-matched* $2 \times 2$ *(speaker* $\times$ *semantics), with* $\pm 1$ *difference standard error.*

The four production models are parameter-matched on this inventory: identical six-parameter set, identical participant hierarchy, and identical constants. Predictive differences are therefore interpreted as reflecting architecture and semantic regime rather than expressive flexibility. Production responses are categorical over the fifteen utterance types; each chain drew 4,000 warmup and 2,000 post-warmup samples (target acceptance probability = 0.85, maximum tree depth = 7). The best-fitting production model is the incremental, context-fixed variant. It captures the qualitative production pattern while remaining deliberately conservative in fit: across the 90 condition $\times$ utterance combinations, observed and posterior-predictive proportions correlate at $r = 0.73$ ($R^2 = 0.53$; Figure 8 shows the by-condition fit and Figure 9 the condition-level correlation).

The largest predictive difference is between speaker architectures. Both incremental models outperform both global models by more than 1,300 ELPD in the matched comparison, and the global models show much weaker condition-level correlations ($r \approx 0.54$).

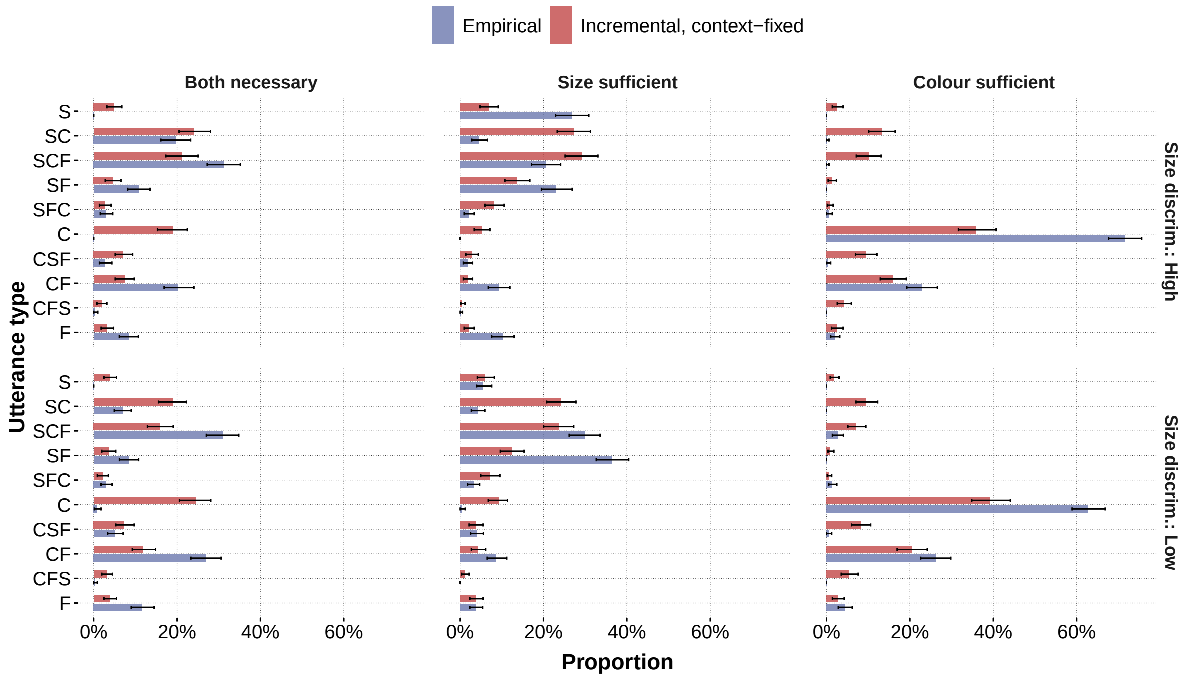



**Figure 8**

*Posterior-predictive utterance-type proportions for the incremental, context-fixed production model. Observed proportions and posterior-predictive model proportions are shown by referential context* $\times$ *size discriminability; error bars are* 95% *bootstrap and posterior-predictive intervals.*

The production comparison therefore supports word-by-word speaker choice over complete-utterance choice as a source of predictive accuracy. This conclusion is about predictive fit; the global models are retained as poor-fitting baselines rather than as sources of substantively interpretable parameter estimates.

Within the incremental pair, the context-fixed model is favoured by PSIS-LOO. The context-fixed model is the best-fitting production model ($\text{ELPD}_{\text{LOO}} = -6{,}367.34$, $p_{\text{LOO}} = 49.01$), followed closely by the context-updating model ($\text{ELPD}_{\text{LOO}} = -6{,}369.32$, $\Delta\text{ELPD} = 1.98$, $\text{dse} = 0.37$). The difference is small relative to the architecture effect but precise at the scale of the incremental-pair comparison, and it is in the direction of the simpler semantic regime. The fitted production comparison therefore provides no evidence

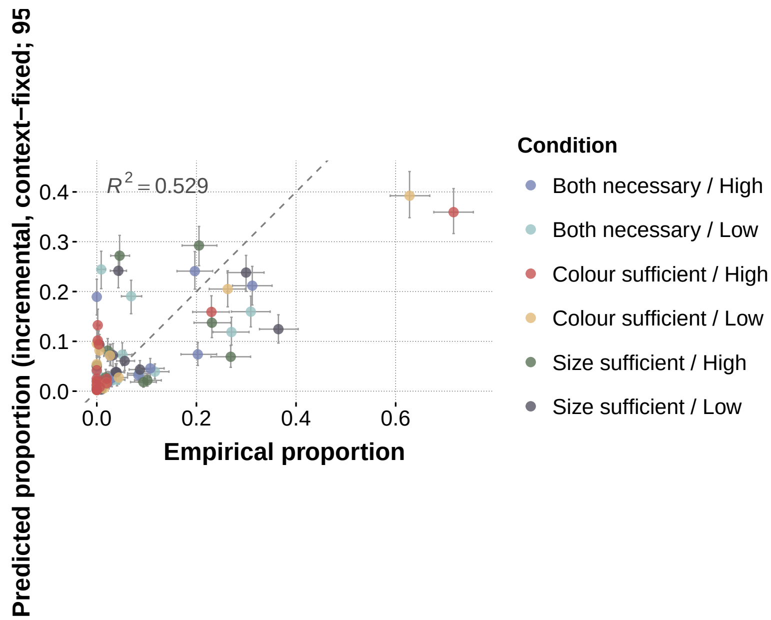



**Figure 9**

*Observed vs. predicted utterance-type proportions across all condition × utterance combinations for the incremental, context-fixed production model; dashed line is identity.*

for an additional predictive benefit from context-updating semantics. The incremental advantage is robust, but posterior-dependent size meanings do not improve out-of-sample prediction once the final production model includes the order-only prior, display-sensitive salience, salience-sensitive stopping, and lapse mixture. The semantic-scale audit in Appendix D clarifies this interpretation: the implemented update only weakly changes target probabilities in the realized displays, so the comparison is most informative about this specific threshold-updating formulation. The same conclusion holds in a robustness check in which the global lapse rate is fixed at the incremental posterior value, eliminating the unstable lapse–rationality trade-off in the global models; the difference-in-differences remains negative ($-2.30$ ELPD), and no model has problematic Pareto-$k$ values.

The posterior parameters of the incremental, context-fixed model are reported in full in Table B1. The posterior for the order-prior weight is clearly positive ($\log \beta \approx 0.59$), but it does not account for the fit by itself. Pragmatic rationality is positive ($\alpha \approx 1.44$), and both salience parameters are clearly identified ($\lambda_{\text{salience}} \approx 1.57$, $\rho_{\text{stop}} \approx 0.49$). The lapse rate is near zero ($\epsilon \approx 0.003$), so the model is not relying on uniform noise to absorb misfit. The participant-level spread is substantial but well behaved ($\tau \approx 0.84$), indicating stable individual variation in pragmatic strength.

### 6.3 Discussion

Comparing the two datasets yields a consistent architecture result. Both datasets favour incremental choice over global utterance evaluation in predictive fit, but they support that conclusion in different response spaces. In the slider data, the two incremental semantic regimes are statistically indistinguishable ($|\Delta\text{ELPD}|/\text{dse} = 0.32$). In the production data, the two incremental regimes are close relative to the architecture effect, but the context-fixed model is favoured by a small and precise difference. Thus the empirical comparison does not support the expected additional benefit of context-updating semantics in production. Context-updating composition still has a defined role in the mechanism space: the simulation asks what it can generate on its own, and it supplies order sensitivity for a global speaker without an order prior. In the fitted production comparison, that role does not translate into best predictive performance once the model includes an order-only prior, visual salience, and incremental choice. The response-space asymmetry is instead reflected in what each task identifies. The binary slider response space is informative about incremental pragmatic evaluation but weakly identifies the sharpness of the colour predicate; the production response space exposes a richer pattern of one-, two-, and three-adjective choices, making the incremental production process more diagnostic.

## 7 Simulation Study

### 7.1 Goals and Design

The fitted model comparison identifies which mechanisms add predictive accuracy, but it does not by itself determine what each mechanism can generate. This leaves an interpretive question for context-updating semantics. In the fitted comparisons, context-updating semantics adds little or no predictive accuracy within the best incremental models. The same mechanism may still be sufficient to produce a baseline size-first asymmetry in a generative model. The simulation addresses this question by applying the same $2 \times 2$ crossing outside the fitted empirical models.

We ask three questions. First, can the crossed models *generate* a size-first ordering preference across a broad range of contexts without being fit to observed responses? Second, when do they generate colour-first reversals? Third, how do speaker architecture and semantic regime combine: does the baseline effect require context-updating semantics, or can it arise from incremental choice alone?

To answer these questions, we ran a parameter-grid simulation over the values listed in Table 3, crossing both speaker models (incremental, global) with both semantic regimes: context-updating (recursive; Equation (7)) and context-fixed (static; Equation (10)). Under context-updating semantics, the size threshold adapts to the literal listener's running posterior at each composition step; under context-fixed semantics, the threshold is computed once from the uniform prior and held fixed. For each parameter combination, 1,000 random contexts were sampled at the specified context size, yielding 12,000,000 simulated trials in total. On each trial, the simulation computed the pragmatic listener's posterior probability of the target referent after the size-first utterance "big blue" and after the colour-first utterance "blue big". Speaker architecture enters through this pragmatic listener: $L_1$ is obtained by Bayesian inversion of either a global or an incremental speaker, so the simulation measures the communicative success of fixed utterances under different speaker architectures rather than fitted utterance frequencies. The simulation estimand is the size-first advantage:

$$\Delta_{\mathrm{SF}} = L_1(o^* \mid \text{"big blue"}) - L_1(o^* \mid \text{"blue big"}).$$

Positive values indicate that the size-first order has higher communicative success; negative values indicate a colour-first advantage. The plotted values are averages of this quantity over sampled contexts and parameter settings within each displayed bin.

The parameter $\sigma_{\text{spread}}$ controls the spread of the size distribution in the sampled context: low values produce low size-discriminability contexts in which size is weakly informative, whereas high values produce high size-discriminability contexts in which size

**Table 3**

*Simulation parameter grid*

| **Parameter** | **Grid values** | **# levels** |
|---|---|---|
| Speaker model | Incremental, global | 2 |
| Semantics | Context-updating, context-fixed | 2 |
| $n_{\text{obj}}$ | 2, 6, 10, 14, 18, 22, 26, 30 | 8 |
| $\sigma_{\text{spread}}$ | 2.0, 7.75, 15.0 | 3 |
| $\nu$ | 0.90, 0.92, 0.94, 0.96, 0.98 | 5 |
| $k$ | 0.10, 0.30, 0.50, 0.70, 0.90 | 5 |
| $w_f$ | 0.1, 0.2, 0.3, 0.5, 0.8 | 5 |
| Random context samples per combination | 1,000 | |
| **Total trials** | **12,000,000** | |

strongly separates the target from its competitors. The remaining parameters ($k$, $w_f$, $\nu$) instantiate the semantic space explored in the simulation. Because the goal is to test the qualitative consequences of semantic regime, the $\nu$ grid focuses on high-reliability determinate predicates, matching the theoretical regime in which colour and form are nearly categorical. The production model, by contrast, uses the empirically calibrated softer colour value $\nu_{\text{colour}} = 0.59$ for predictive fitting. The simulation is therefore a theoretical parameter-grid study rather than a posterior-predictive simulation of the fitted production model. The individual effects of the semantic grid parameters are reported in Appendix E.

### 7.2 Results

Figure 10 shows the size-first advantage as a function of context size and contextual spread, faceted by speaker model (columns) and semantic regime (rows).

The simulation yields three results that qualify the model-comparison interpretation.

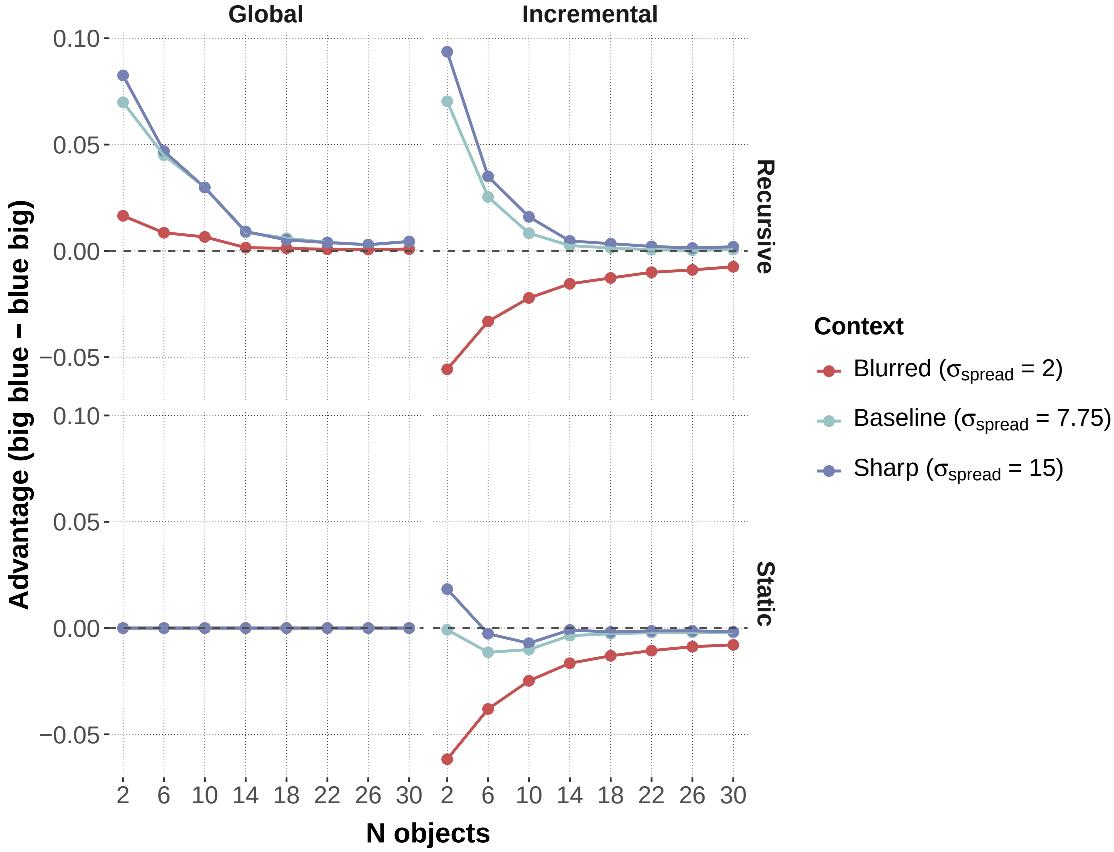



**Figure 10**

*Simulation size-first advantage* $\Delta_{SF} = L_1(o^* \mid \text{“big blue”}) - L_1(o^* \mid \text{“blue big”})$ *as a function of context size* ($n_{obj}$) *and contextual spread* ($\sigma_{spread}$)*, faceted by speaker model (columns) and semantic regime (rows). Positive values indicate higher communicative success for the size-first order.*

First, context-updating semantics can generate a size-first advantage without incremental speaker choice. Under context-updating semantics, both speaker models generate a size-first advantage across most of the parameter space. A global speaker with context-sensitive composition therefore derives the canonical ordering asymmetry without an incremental production policy. This result is consistent with the subjectivity literature: once semantic values are evaluated over progressively restricted comparison classes, less noisy inner modification can produce the familiar ordering direction (Franke et al., 2019; Scontras et al., 2020, 2019).

Second, context-fixed semantics removes this global speaker advantage. Under context-fixed semantics, the global speaker produced $\Delta = 0.000$ across the parameter grid. When the size threshold does not update during composition, the literal listener assigns identical posteriors to both orderings, making them informationally equivalent. The global speaker's baseline advantage therefore comes from order-sensitive threshold adaptation rather than from utterance-level choice.

Third, incremental choice changes the shape of the ordering pattern. The incremental speaker strengthens the size-first advantage under context-updating semantics in much of the parameter space, and it also produces colour-first preferences in low size-discriminability, low-cardinality contexts. Under context-fixed semantics, the incremental speaker defaults slightly to colour-first ($\Delta = -0.009$), driven by the chain-rule asymmetry that favours "blue" as the opening token when size is insufficiently diagnostic. The simulation therefore separates two sources of order sensitivity: context-updating composition can provide a baseline asymmetry, and incremental choice determines how adjective order responds to the relative informativeness of each adjective at the point where it is selected.

## 8 General Discussion

The present paper asked how stable adjective-ordering expectations are combined with referential informativeness during communication. The empirical results show that both pressures are visible. Across tasks, participants retained a size-first baseline, especially in slider judgments, and shifted their responses with the property that best identified the target in the current display. Production made this context sensitivity more categorical and revealed an additional length effect: speakers used more redundant adjective material when size was harder to discriminate. This length effect is consistent with work on communicative efficiency and redundant modification, where additional material can support reference by helping listeners resolve the target efficiently (Degen et al., 2020; Rubio-Fernández, 2016; Rubio-Fernandez, 2019).

The model comparison localizes the strongest predictive gain in speaker architecture. Incremental speakers outperformed global speakers in both response formats, indicating that much of the context sensitivity in the observed data is captured by word-by-word pragmatic choice. The semantic-regime contrast was weaker in the fitted models. Context-updating and context-fixed semantics were indistinguishable in the slider task, and the context-fixed model was slightly favoured in production once the order prior, display-sensitive salience, stopping pressure, and lapse mixture were included.

The simulation clarifies how to interpret this result. Context-updating composition can generate a size-first advantage without incremental choice because the comparison class changes over the course of semantic composition. This gives the subjectivity-based mechanism a clear generative role. The fitted experiments show a different predictive division of labor: in these data, incremental pragmatic choice and production-level salience account for more variation than posterior-dependent size meanings add on top of the shared model components.

The size-discriminability manipulation supports the same cautious interpretation. Lower visual size reliability did not produce a uniform increase in size-first ordering, so the display-level proxy prediction for P3 receives limited support. At the same time, size discriminability mattered for production length and interacted weakly with referential context in judgment. These patterns place perceptual reliability mainly in salience and effort-sensitive adjective selection, with only limited evidence for a direct monotonic shift in order preference.

The study has several limits. The experiments focused on German prenominal adjective order, and the main analyses focused on the size–colour subset. The size-discriminability manipulation changed visual reliability in the display, so it provides an indirect test of subjectivity-linked predictions. The production task also constrained participants to the adjective slot before a fixed noun, which improved coding but left aside phenomena that would appear in unconstrained referring expressions. Future work could combine direct manipulations of lexical subjectivity with freer production tasks and languages with different adjective-ordering systems.

Taken together, the results support an account in which adjective order is shaped by stable ordering expectations, scene-specific informativeness, perceptual salience, and incremental choice. Subjectivity-based composition explains how canonical orders can arise from meaning, and the fitted experiments show that actual referential responses are best predicted when the model also includes incremental pragmatic selection and production-level salience.

## 9 Data Availability Statement

Data, analysis scripts, model code, and figure-generation scripts will be made available in an anonymized repository upon submission. The repository is intended to contain the cleaned experimental data, preprocessing scripts, NumPyro model code, posterior summaries, and manuscript figure scripts needed to reproduce the reported analyses.

## 10 Author Note

Acknowledgments, funding information, author contributions, and competing-interest statements will be finalized with the author team before journal submission.

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# Appendix A

## Methods and Coding Transparency

Table A1 summarizes the participant exclusions and the analysis tables used in the reported manuscript analyses. The per-criterion exclusion counts are not mutually exclusive, so the unique number of excluded participants is smaller than the sum of the criterion counts. For Experiment 2, the broad preprocessed production table contains all three critical adjective-pair families. The production model comparison reported in Section 6.2 uses the size–colour subset with annotated responses, yielding 3,196 observations over the 15 utterance categories.

**Table A1**

*Participant exclusions and analysis-table sizes. Criterion counts overlap within experiment.*

| Experiment | Starting and final $N$ | Participant exclusions | Analysis table used in manuscript |
|---|---|---|---|
| Slider ratings | 150 → 134 | 8 with more than three missing trial-list entries; 8 with extreme completion time; 2 with more than three control-item failures; 6 with more than three "none fits" responses on experimental items; 16 unique participants excluded. | Broad cleaned critical table: 10,606 observations. Reported size–colour slider subset: 3,485 observations. |
| Constrained production | 149 → 113 | 8 with extreme completion time; 25 with more than 80% identical responses; 7 with more than 10% missing or unannotatable responses; 36 unique participants excluded. | Broad preprocessed critical table: 9,639 rows. Reported size–colour production model and figures: 3,196 annotated observations. |

The slider-rating materials used the fixed adjective vocabulary in Table A2. The raw stimulus table encodes colours and forms with compact feature labels; the interface displayed the corresponding German adjective forms in phrases such as *den großen blauen Aufkleber*.

**Table A2**

*Adjective vocabulary in the slider-rating item descriptions.*

| Property class | Raw stimulus values | German adjective forms displayed in the item text |
|---|---|---|
| Size | Numeric size levels `1-6`, `9`, `10` | *großen*, *kleinen* |
| Colour | `blue`, `orange`, `grey`, `brown`, `green`, `black` | *blauen*, *orangen*, *grauen*, *braunen*, *grünen*, *schwarzen* |
| Form | `circle`, `diamond`, `heart`, `quadrat`, `star`, `triangle` | *runden*, *diamantförmigen*, *herzförmigen*, *quadratischen*, *sternförmigen*, *dreieckigen* |

The preprocessed modelling table recodes the determinate contrasts to binary feature values, but this recoding is separate from the lexical forms shown to participants.

Production responses were coded into the 15 ordered adjective categories listed in Table A3. The data files use `D` for the size or dimension adjective, while the model exposition uses S; the two notations refer to the same adjective class. Misspellings and transparent lexical alternatives were normalized when their semantic class was unambiguous. Spatial descriptions, comparative or ordinal size expressions, and responses that could not be assigned to a target adjective class were excluded from the production model comparison.

**Table A3**

*Production coding dictionary. In the data files, `D` corresponds to the size adjective S; `C` and `F` correspond to colour and form.*

| Length | Codes |
|---|---|
| One adjective | `D`, `C`, `F` |
| Two adjectives | `DC`, `DF`, `CD`, `CF`, `FD`, `FC` |
| Three adjectives | `DCF`, `DFC`, `CDF`, `CFD`, `FDC`, `FCD` |

The descriptive analyses and the Bayesian model comparisons address different inferential questions. The descriptive regressions test the qualitative empirical patterns in the observed responses, whereas the Bayesian models compare predictive mechanisms under response-appropriate likelihoods. For the reported size–colour slider analysis, the descriptive model was a linear mixed-effects model of centred slider rating with fixed effects of referential context, size discriminability, and their interaction, plus random intercepts for participant and item:

$$y_{\text{slider}} \sim \text{context} \times \text{size discriminability} + (1 \mid \text{participant}) + (1 \mid \text{item}).$$

For production size-initiality, the descriptive model was a logistic mixed-effects model with a binary outcome indicating whether the coded response began with the size adjective:

$$y_{\text{size-initial}} \sim \text{context} \times \text{size discriminability} + (1 \mid \text{participant}) + (1 \mid \text{item}).$$

For production over-informativity, the descriptive model used the same fixed and random structure with a binary outcome indicating a three-adjective response in both-necessary contexts or a multi-adjective response in one-property-sufficient contexts:

$$y_{\text{over-informative}} \sim \text{context} \times \text{size discriminability} + (1 \mid \text{participant}) + (1 \mid \text{item}).$$

## Appendix B

## Production Model Details

Table B1 lists the population parameters of the advocated production model (Section 6.2), their priors, and their posterior summaries. The participant hierarchy adds 113 latent rationality offsets drawn from a zero-mean normal distribution with standard deviation $\tau$. The fixed constants are $\nu_{\text{colour}} = 0.59$, $\nu_{\text{form}} = 0.50$, $k = 0.50$, and $w_f = 0.69$.

**Table B1**

*Population parameters of the advocated production model: prior and posterior mean with 94% HDI.*

| Parameter | Role | Prior | Posterior |
|---|---|---|---|
| $\alpha$ | pragmatic rationality | HalfNormal(3) | 1.44 [1.17, 1.68] |
| $\log \beta$ | order-only LM-prior weight | $\text{Normal}(0, 0.35)$ | 0.59 [0.54, 0.65] |
| $\lambda_{\text{salience}}$ | local boost for visually salient features | HalfNormal(1) | 1.57 [1.50, 1.65] |
| $\rho_{\text{stop}}$ | cost for continuing after a salient feature | HalfNormal(0.75) | 0.49 [0.41, 0.59] |
| $\epsilon$ | lapse / uniform-noise rate | $\text{Beta}(1, 50)$ | 0.003 [0.000, 0.009] |
| $\tau$ | SD of participant rationality offsets | HalfNormal(0.15) | 0.84 [0.69, 0.99] |

# Appendix C

## Model Diagnostics and Production Robustness

Table C1 summarizes the available diagnostics for the main $2 \times 2$ comparisons. For the slider models, manuscript-facing summaries contain PSIS-LOO warnings and maximum split-$\hat{R}$, but they do not archive Pareto-$k$ counts or ESS/divergence summaries. For the production models, the incremental models have clean MCMC diagnostics and no problematic Pareto-$k$ observations; the global models fit much worse and have weak convergence, so their parameters are not interpreted substantively.

**Table C1**

*Model-comparison diagnostics. ΔELPD and dse are relative to the best model within dataset. Pareto-k counts are the number of observations with $k \geq .7$ when archived.*

| Model | Predictive summary | Diagnostics | Interpretation |
|---|---|---|---|
| **Slider**: incremental, updating | ELPD −2774.1; $p_{\mathrm{LOO}}$ = 64.8; Δ = 0.0; dse – | PSIS warning yes; max $\hat{R}$ = 1.006; Pareto-$k$ n/a; divergences n/a | Best slider model; incremental comparison trusted with PSIS warning noted. |
| **Slider**: incremental, fixed | ELPD −2774.1; $p_{\mathrm{LOO}}$ = 65.1; Δ = 0.06; dse 0.18 | PSIS warning yes; max $\hat{R}$ = 1.007; Pareto-$k$ n/a; divergences n/a | Statistically indistinguishable from updating. |
| **Slider**: global, updating | ELPD −4249.2; $p_{\mathrm{LOO}}$ = 1425.0; Δ = 1475.1; dse 65.9 | PSIS warning yes; max $\hat{R}$ = 1.845; Pareto-$k$ n/a; divergences n/a | Poor predictive baseline; parameters not interpreted. |
| **Slider**: global, fixed | ELPD −7562.2; $p_{\mathrm{LOO}}$ = 4372.0; Δ = 4788.1; dse 127.2 | PSIS warning yes; max $\hat{R}$ = 3.098; Pareto-$k$ n/a; divergences n/a | Poor predictive baseline; parameters not interpreted. |
| **Production**: incremental, fixed | ELPD −6367.3; $p_{\mathrm{LOO}}$ = 49.0; Δ = 0.0; dse – | PSIS warning no; $k \geq .7$: 0; max $\hat{R}$ = 1.00; divergences 0 | Best production model; main inferential reference. |
| **Production**: incremental, updating | ELPD −6369.3; $p_{\mathrm{LOO}}$ = 48.6; Δ = 1.98; dse 0.37 | PSIS warning no; $k \geq .7$: 0; max $\hat{R}$ = 1.00; divergences 0 | Clean diagnostics; close but lower than fixed semantics. |
| **Production**: global, updating | ELPD −7730.7; $p_{\mathrm{LOO}}$ = 615.5; Δ = 1363.4; dse 33.3 | PSIS warning yes; $k \geq .7$: 2; max $\hat{R}$ = 1.53; divergences 0 | Poor predictive baseline with weak convergence. |
| **Production**: global, fixed | ELPD −7731.1; $p_{\mathrm{LOO}}$ = 615.8; Δ = 1363.7; dse 33.4 | PSIS warning yes; $k \geq .7$: 1; max $\hat{R}$ = 1.54; divergences 0 | Poor predictive baseline with weak convergence. |

The main production comparison uses PSIS-LOO on the matched six-parameter $2 \times 2$. The two incremental models have clean diagnostics (maximum split-$\hat{R} = 1.00$, no divergences, and no Pareto-$k \geq .7$ observations). The global models are much worse fitting. Because global utterance-level choice can trade off strongly with the lapse term, we also checked a version in which the global lapse rate was fixed at $\epsilon = .003$, the posterior value of the advocated incremental model. This check leaves the substantive ordering of the models unchanged: incremental context-fixed remains best, incremental context-updating remains second, and the difference-in-differences for the speaker $\times$ semantics contrast remains negative rather than positive ($-2.30$ ELPD). The fixed-lapse global check removes problematic Pareto-$k$ values; global-chain convergence also improves substantially, though the global models remain less stable than the advocated incremental models (maximum split-$\hat{R} = 1.07$ for global context-updating and 1.05 for global context-fixed).

# Appendix D

## Production Model Development and Fixed-Constant Provenance

Tables D1 and D2 summarize the provenance of the fixed constants and production-model components used in the final matched comparison. These choices were not varied across the four models of the final $2 \times 2$. They therefore define the common production speaker against which the architecture and semantic-regime contrasts are tested.

**Table D1**

*Fixed constants used in the final matched comparison.*

| Constant | Value | Provenance or check | Claim status |
|---|---|---|---|
| Slider $\nu$ | 0.80 | Minimal slider-speaker specification for the size–colour slider data; free-$\nu$ audit reported in Appendix F. | Architecture conclusion unchanged; bias estimate is conditional on this fixed value. |
| Production $\nu_{\text{colour}}$ | 0.59 | Free-colour-semantic-value diagnostic on the size–colour production data; value re-fixed for cleaner matched fits. | Defines common production semantics; not itself a $2 \times 2$ contrast. |
| Production $\nu_{\text{form}}$ | 0.50 | Fixed soft determinate-feature constant used to retain form as a possible redundant modifier. | Form is retained for response-space coverage, not as the main mechanistic contrast. |
| Production $k$ | 0.50 | Mid-range threshold default; simulation sweeps $k$ in Appendix E. | Held fixed so models differ only in architecture and semantic regime. |
| Production $w_f$ | 0.69 | Previous free-$w_f$ diagnostic median from production model development; simulation sweeps $w_f$ in Appendix E. | Held fixed in the production comparison. |

The archived constant name for the production blur weight is `WF_FIXED_ITER11_MEDIAN`.

We also audited the scale of the semantic-regime manipulation on the 162 unique size–colour displays. Under the production constants ($\nu_{\text{colour}} = 0.59$, $k = 0.50$, $w_f = 0.69$), recomputing the size threshold after a colour update changed the target's graded size value by $< .001$ on average and changed the literal listener's target probability for SC by $< .001$. Under the slider constants ($\nu = 0.80$, $k = 0.50$, $w_f = 1.0$), the mean threshold shift was larger, but the corresponding listener-probability change also remained below 0.001. The audit explains why the fitted semantic-regime contrast is conservative in these materials: the current mid-range threshold and ANS-style slack make posterior-dependent size meaning a weak perturbation unless the colour predicate is sharper and the size-membership function is less diffuse. A full-range max–min threshold did not strengthen the contrast in this stimulus set because the relevant colour-conditioned subsets usually preserved the same size extrema.

**Table D2**

*Production-component provenance for the final matched comparison.*

| Component | Form | Provenance or check | Claim status |
|---|---|---|---|
| Order-only LM prior | Residual score $\Omega(u)$ | German GPT-2 surprisal residualized within unordered content sets; weight $\beta$ inferred in every final model. | Retained; cannot explain length or content choice by construction. |
| Soft visual salience | Centered display score $z_a$ | Fixed base vector plus display adjustment; no referential-context dummy; $\lambda_{\text{salience}}$ inferred. | Retained as non-pragmatic feature prominence. |
| Salience-sensitive stopping | Continuation cost after salient words | Production model-development check on size–colour data; improves PSIS-LOO over the no-length variant by 52.57 ELPD (dse = 10.07). | Retained as short-description pressure. |
| Uncertainty-gated length term | Removed | Exploratory production check on size–colour data showed no clear benefit over the model without the term ($\Delta$ELPD = 0.49, dse = 0.88), and the posterior included zero. | Not part of the final production model. |
| Regularized hierarchy | Participant offsets with conservative prior | Model-development check; slightly lower ELPD than the unregularized salience-stop fit ($\Delta\text{ELPD} \approx 2.16$, dse = 1.15) while reducing $p_{\text{LOO}}$. | Used for final matched comparison. |

Table D3 summarizes the main production-model development checks. These checks were used to keep the final production model tied to independently motivated components while avoiding condition-specific residual parameters. The final $2 \times 2$ comparison uses the regularized salience-stop model because it retains the qualitative fit of the best exploratory variant while reducing effective complexity.

**Table D3**

*Production-model development checks motivating the final six-parameter inventory.*

| Component or choice | Motivation | Check | Decision |
|---|---|---|---|
| Order-only LM prior | Stable order expectations without allowing the LM prior to explain content choice or utterance length. | LM scores residualized within unordered content sets, so the prior distinguishes SC from CS but not short from long descriptions. | Retained as $\Omega(u)$ with inferred weight $\beta$. |
| Soft visual salience | Feature prominence is independent of pragmatic information gain and should vary with the display. | Salience is context-dependent and centered within trial; no referential-context dummy is used. | Retained as local first-word boost $\lambda_{\text{salience}} z_a$. |
| Salience-sensitive stopping | Highly salient sufficient features should support short descriptions. | Adding the stopping term improves PSIS-LOO over the salience-only/no-length variant ($\Delta$ELPD = 52.57, dse = 10.07). | Retained as the continuation cost in Equation (21). |
| Uncertainty-gated length term | Blurry size could in principle motivate longer descriptions. | Initial comparison showed no clear benefit over the model without the term ($\Delta$ELPD = 0.49, dse = 0.88), and the posterior included zero. | Removed from the final model. |
| Regularized participant hierarchy | Participant variation is required, but the final architecture comparison should be conservative about effective complexity. | Regularization slightly lowers ELPD relative to the unregularized salience-stop fit ($\Delta$ELPD $\approx$ 2.16, dse = 1.15) while reducing $p_{\text{LOO}}$ from about 56.4 to 48.9. | Used for the final 2×2 comparison. |

## Appendix E

## Simulation Parameter Sweeps

Figures E1 to E3 show how the three fixed semantic parameters affect communicative success across the two speaker models.

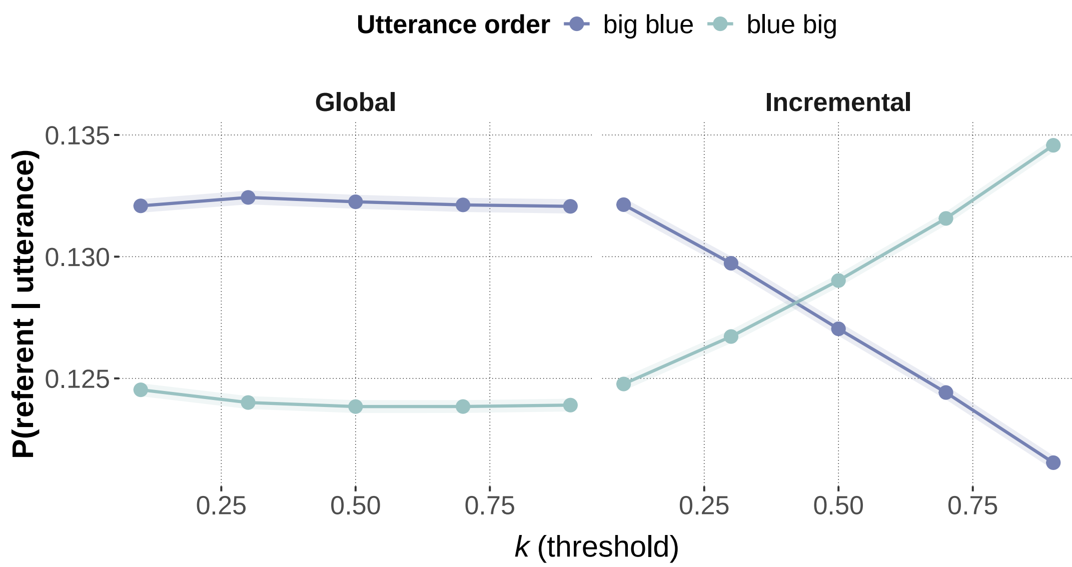


**Figure E1**

*Effect of the threshold parameter k on communicative success in the simulation.*

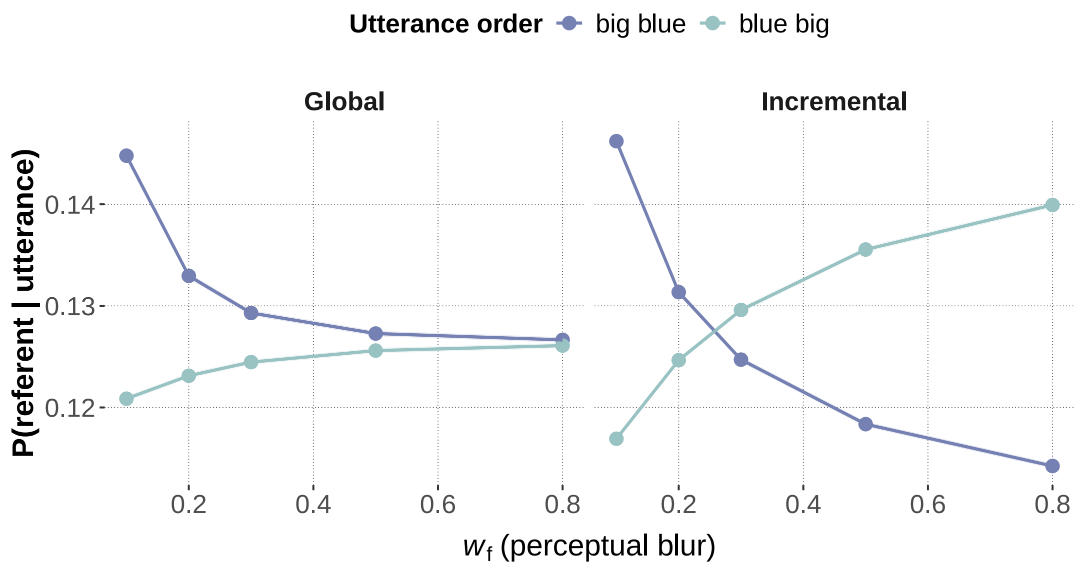


**Figure E2**

*Effect of perceptual blur $w_f$ on communicative success in the simulation.*

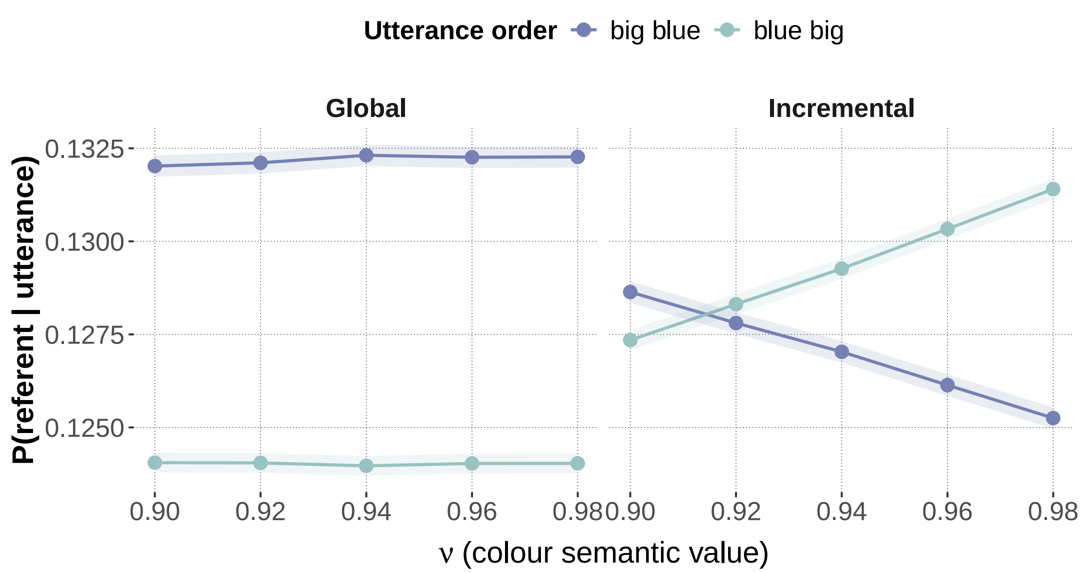


**Figure E3**

*Effect of the colour/form semantic value $\nu$ on communicative success in the simulation.*

# Appendix F

## Slider Colour-Reliability Audit

The slider-rating model fixes the colour and form reliability $\nu$ at 0.80 (Section 6). Production uses the softer colour value $\nu_{\text{colour}} = 0.59$ in the final principled speaker (Section 6.2); for parity we apply the same kind of sensitivity audit to the slider's incremental models. The audit samples $\nu \sim \text{Uniform}(0.5, 0.999)$ within the model and recomputes the literal listeners at each MCMC step, while keeping every other component of the incremental hierarchical speaker unchanged. Table F1 reports the resulting posteriors alongside the originally fixed-$\nu$ fits.

**Table F1**

*Slider colour-reliability audit. Fixed-$\nu$ rows use $\nu = 0.80$; free-$\nu$ rows sample $\nu \sim \text{Uniform}(0.5, 0.999)$ and recompute literal listeners each step. "ELPD" is PSIS-LOO; $\Delta$ELPD and dse are relative to the incremental, context-updating fixed-$\nu$ model. All four models use 4 chains $\times$ 500 warmup $\times$ 500 samples; maximum split-$\hat{R}$ rounds to 1.01 for the free-$\nu$ models.*

| Model | ELPD | ΔELPD | dse | $\alpha$ (sd) | $b$ (sd) | $\nu$ (94% HDI) |
|---|---|---|---|---|---|---|
| **Incremental, updating, fixed** $\nu = 0.80$ | **−2,774.1** | **0.0** | — | **0.69 (0.06)** | **0.58 (0.06)** | — |
| Incremental, fixed, fixed $\nu = 0.80$ | −2,774.1 | −0.06 | 0.18 | 0.69 (0.06) | 0.58 (0.06) | — |
| Incremental, updating, free $\nu$ | −2,775.0 | −0.89 | 1.08 | 1.53 (0.61) | 0.41 (0.15) | 0.63 [0.50, 0.82] |
| Incremental, fixed, free $\nu$ | −2,774.3 | −0.23 | 1.00 | 1.48 (0.55) | 0.42 (0.13) | 0.63 [0.51, 0.79] |

Three points stand out. First, $\nu$ is weakly identified in the slider data: the posterior places substantial mass near the lower prior boundary at 0.5 and only just excludes the fixed value at the upper end, in contrast to the tight production posterior 0.59 [0.56, 0.62]. Second, the four models are statistically indistinguishable in PSIS-LOO ($|\Delta\text{ELPD}|/\text{dse} < 1$

throughout), so out-of-sample fit cannot decide between sharper and softer colour predicates in this response space. Third, under the freed posterior the rationality parameter $\alpha$ moves to roughly 1.5 (from 0.69) and the bias intercept $b$ moves below 0.5 (from 0.58), because $\nu$ and the other parameters trade off: a softer colour predicate forces a stronger rationality term and reduces the residual size-first bias absorbed by $b$. We therefore retain the fixed $\nu = 0.80$ for the slider model in the main text and treat $\alpha$ and $b$ at that specification as the reported quantities, while flagging the conditional status of the $b > 0.5$ finding (Section 6, Slider-Rating Study). The cross-dataset asymmetry — production requires a softer colour predicate in the final principled speaker, while the slider data do not strongly identify $\nu$ — is itself diagnostic of the response-space sensitivity argued for in the Cross-Dataset Summary.